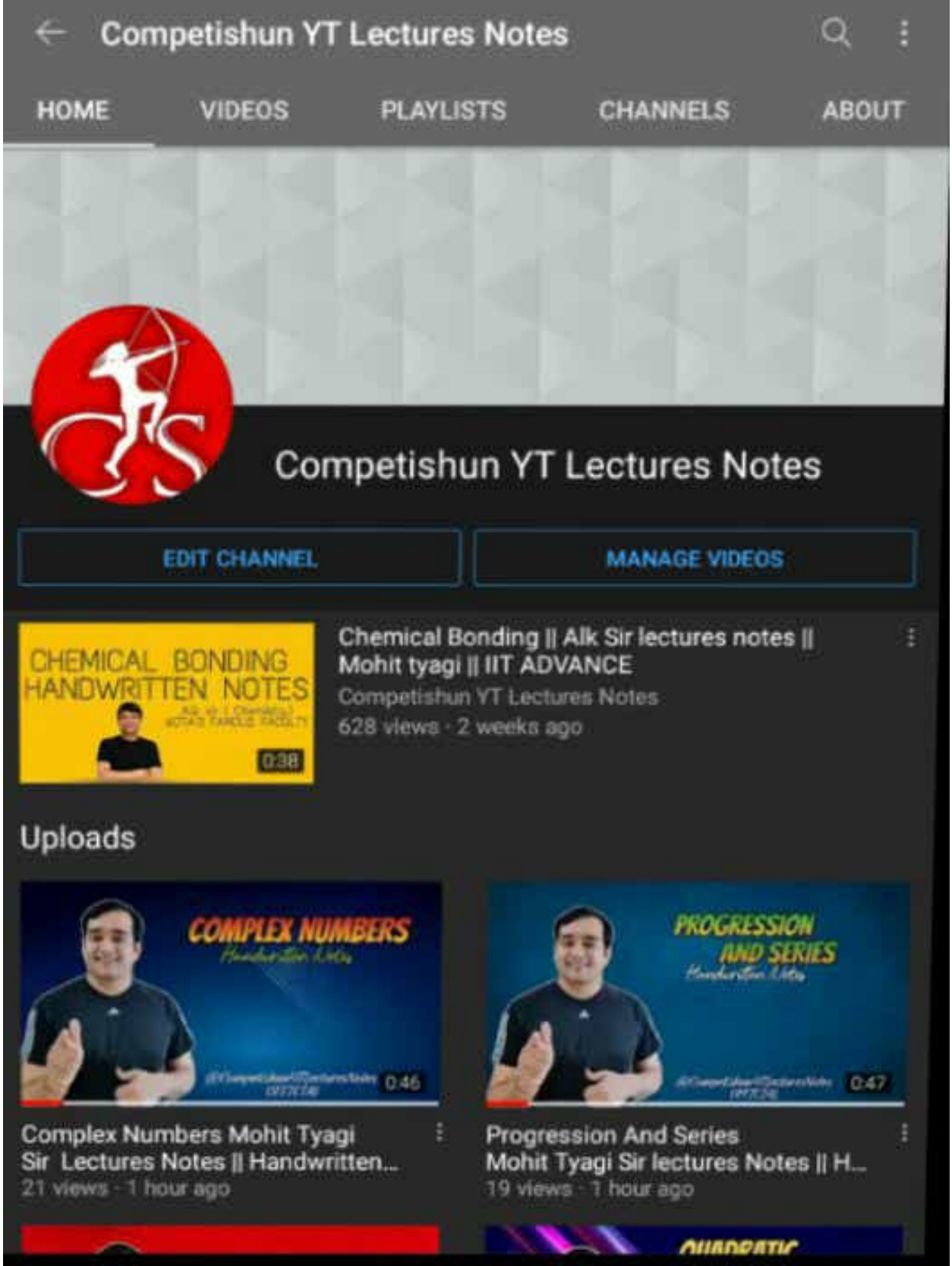


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ORGANIC REACTION MECHANISM -I (ORM-I)

(I) Introduction of reaction mechanism
- [A] Types of bond fission
- [B] Types of solvents
- [C] Attacking reagents
- [D] Nucleophilicity
- (E) Leaving group ability
- (F) Types of org reactions.

(II) Acid-Base reaction :-

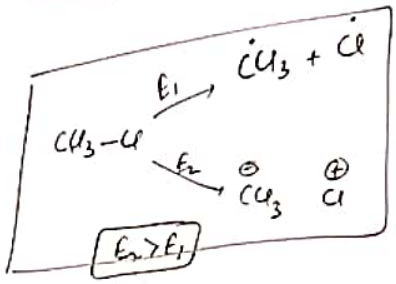
(III) Nucleophilic Addition reaction :-

(IV) Acyl SN^2 Reaction = SN^2Th reaction :-

ORGANIC REACTION MECHANISM -I (ORM-I)

(I) Introduction of reaction mechanism-

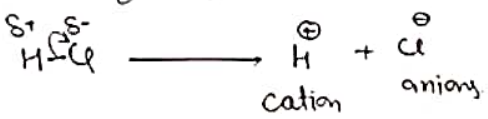
(I.A) Types of Bond Fission (Breaking of bonds)



Homolytic fission



Heterolytic fission



Favourable conditions.

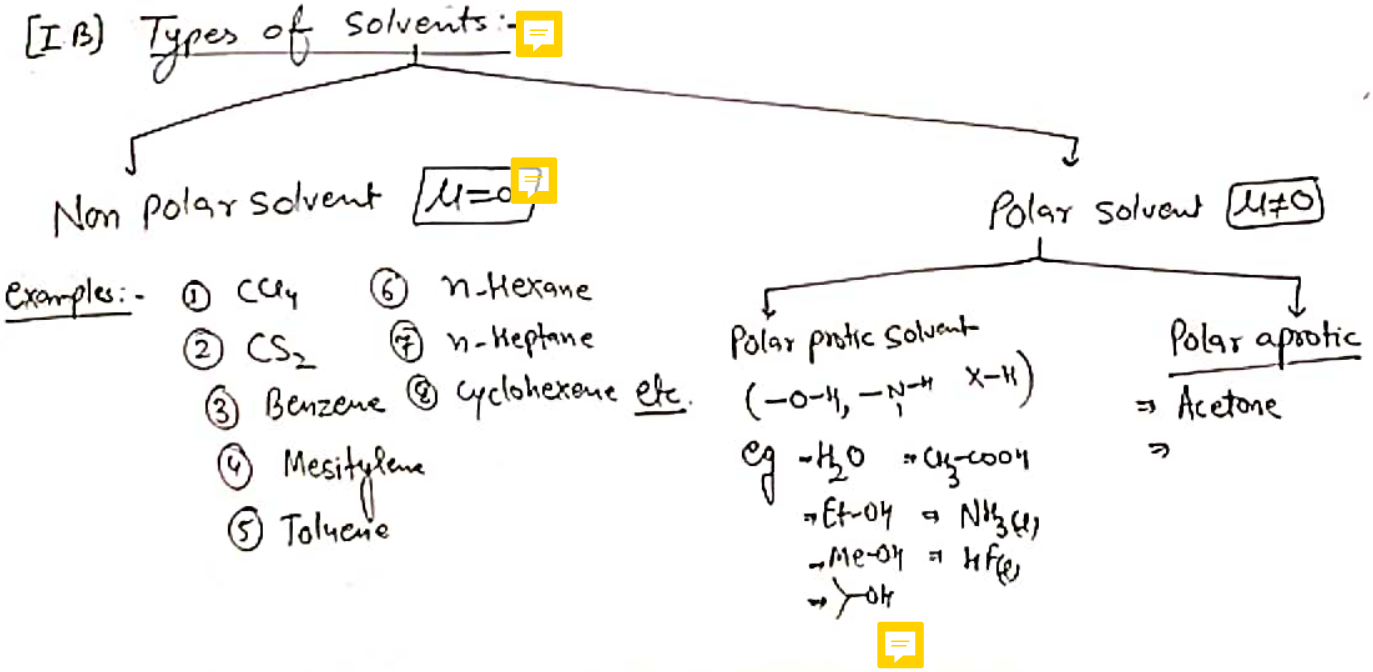
- ⇒ Non polar bond / approx. non polar bond
- ⇒ in gas phase or in non polar solvents
- ⇒ High temp. / Electricity / Free radical / hν / Peroxide, etc

- ⇒ high temp.
- ⇒ in presence of Polar solvent
- ⇒

ORGANIC REACTION MECHANISM -I (ORM-I)

(I) Introduction of reaction mechanism-

[I B] Types of Solvents :-



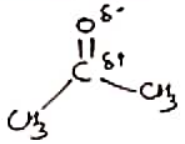
ORGANIC REACTION MECHANISM -I (ORM-I)

(I) Introduction of reaction mechanism–

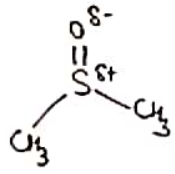
[I.B] Types of Solvents :-

Examples of polar aprotic solvents:

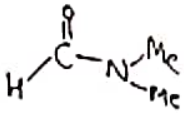
eg ① Acetone



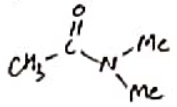
② DMSO
Dimethylsulphoxide



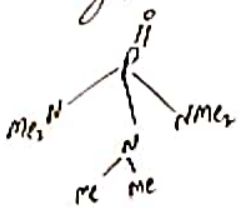
eg ③ DMF = N,N Dimethyl formamide



eg DMA = N,N-Dimethylacetamide



eg HMPA/ HMTA Hexamethylphosphamide



ORGANIC REACTION MECHANISM - I

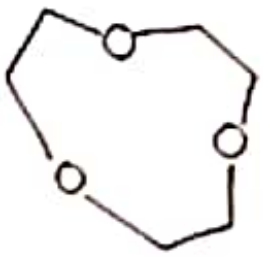
(I) Introduction of reaction mechanism-

[I.B] Types of Solvents :-

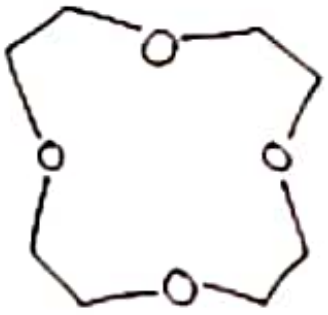
Examples of polar aprotic solvents

eg: Crown ethers: · cyclic poly ethers

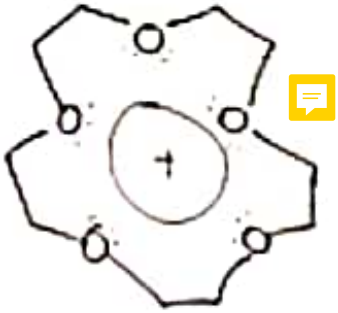
$$\left[-CH_2-CH_2-O\right]_n$$



9-Crown-3



12-Crown-4

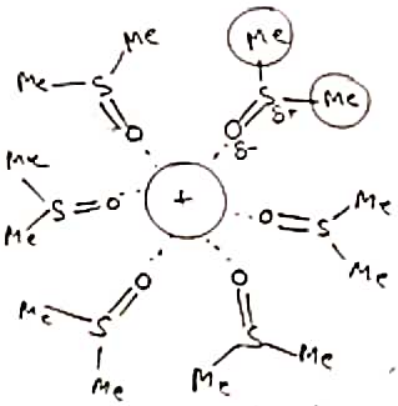


15-Crown-5

ORGANIC REACTION MECHANISM -I (ORM-I)

(I) Introduction of reaction mechanism–

[I B] Types of Solvents :-



Solvation of cation in DMSO



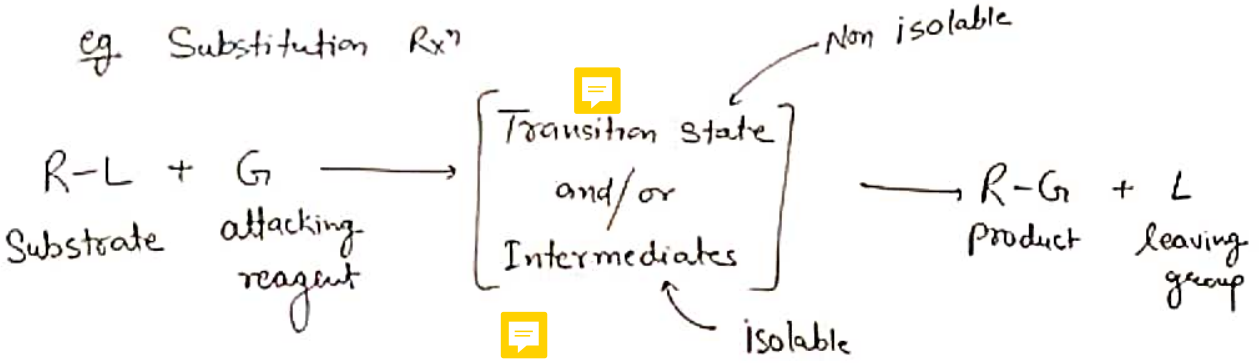
Solvation

	Polar protic solvent	Polar aprotic solvent
Cation	✓	✓
anions	✓	✗

ORGANIC REACTION MECHANISM-I (ORM-I)

I.c Types of attacking Reagents:-

eg. Substitution Rxn



ORGANIC REACTION MECHANISM-I (ORM-I)

I.C Types of attacking Reagents:-

Nucleophiles ($Nu^{\ominus}$)
(Nucleus loving species)

- Lone pair containing, e^- rich species.
- Can attack on e^- dificient position of substrate.

Electrophile ($E^{\oplus}$)
(Electron loving species)

- Vacant orbital containing, e^- dificient species.
- attack on e^- rich position of substrate.

I.c Types of attacking Reagents:-

Nucleophile

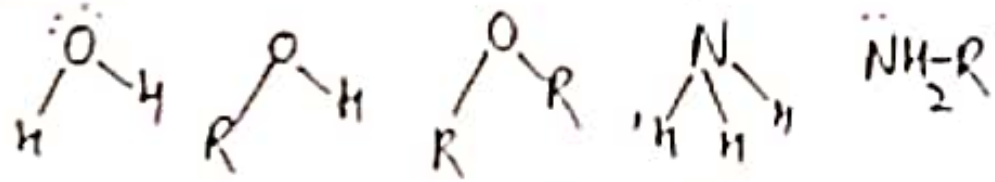
Examples

(a) -ve Charged Nucleophiles.

$H^{\ominus}$, $R^{\ominus}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{O}R$, $\overset{\ominus}{N}H_2$, $\overset{\ominus}{X}$, $\overset{\ominus}{S}H$, $\overset{\ominus}{S}R$

$\overset{\ominus}{C}N$, $R-\overset{\ominus}{\underset{\underset{O}{\|}}{C}}-O^{\ominus}$, $\overset{\ominus}{N}_3$, $Ph-O^{\ominus}$ etc

(b) Neutral Nucleophiles:-

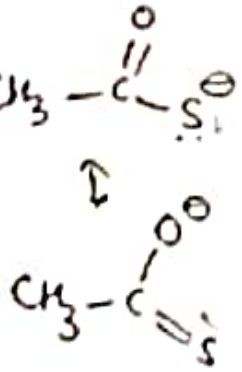
→ O 

R_2NH, R_3N, H_2S, $R-SH$ $R-S-R$  etc

REACTION MECHANISM - I (ORM - I)

Reagents :-

→ (more than one donar atoms)

→ 

Electrophiles

Examples

① +ve charged electrophiles:

$H^{\oplus}$, $R^{\oplus}$, $NO_2^{\oplus}$, $HSO_3^{\oplus}$, $R-C^{\oplus}=O$ (acylinium ion), $X^{\oplus}$ etc

→ ($NH_4^{\oplus}$, $H_3O^{\oplus}$) - Not electrophile
- No vacant orbitals

② Neutral electrophiles :-

⇒ Vacant p orbital containing

BX_3, B_2H_6 (Diborane), $AlCl_3$

$\ddot{C}l_2$ (carbene), $\ddot{N}H$ (Nitrene)

ORGANIC REACTION MECHANISM - I (ORM-I)

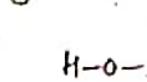
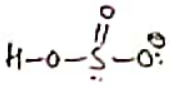
I.c Types of attacking Reagents:-

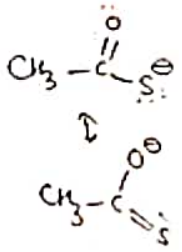
Nucleophile



Examples

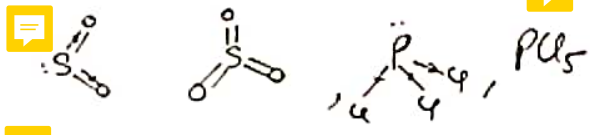
\# Ambident Nucleophile:- (more than one donar atoms)

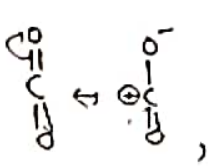
- $\overset{\ominus}{C}N = (\ddot{C}\equiv\ddot{N})^{\ominus}$
- $NO_2^{\ominus} =$ 
- $HSO_3^{\ominus}$ 

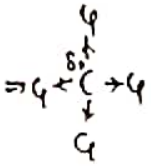
- 

Electrophiles

- Vacant d orbital containing.



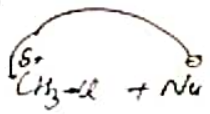
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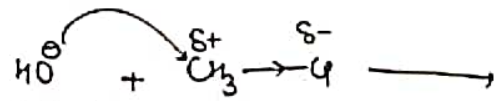
Ald. / Ketone

ORGANIC REACTION MECHANISM-I (ORM-I)

I.D :- **Nucleophilicity**:- It is tendency of nucleophile to attack on e^- deficient position of substrate. (Kinetic property)



Basicity:- It is e^- pair donating tendency to $H^{\oplus}$ (Thermodynamic property)



Nucleophile



Base

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 1.- -ve charged species are stronger nucleophile as well as stronger base with respect to corresponding neutral species.

eg $\overset{\ominus}{O}H > H_2\ddot{O}:$ Nucleophilicity & Basicity.

eg $:\overset{\ominus}{N}H_2 > \ddot{N}H_3$ " "

eg $Ph\overset{\ominus}{O} > Ph-\ddot{O}H$ " "

eg $CH_3-COO^- > CH_3-\overset{\overset{O}{\|}}{C}-\ddot{O}H$ " "

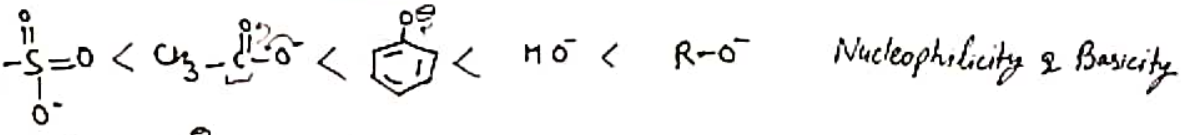
eg $:\overset{\ominus}{S}H > H_2\ddot{S}$ Nucleophilicity & Basicity

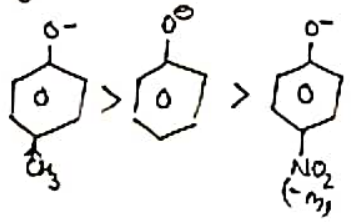
ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 2. When donor atoms are same in diffirent species then order of nucleophilicity and order of basicity are follow same order.

Basic strength $\propto \frac{1}{\text{Stability}}$

Eg 1  Nucleophilicity & Basicity

Eg 2  Nucleophilicity & Basicity

ORGANIC REACTION MECHANISM-I (ORM-I)

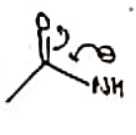
Rules for comparison:-

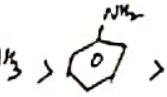
Rule 2. When donor atoms are same in different species then order of nucleophilicity and order of basicity are follow same order.

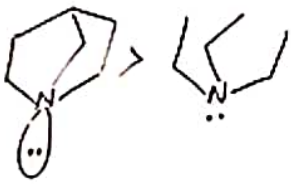
$$\text{Basic strength} \propto \frac{1}{\text{Stability}}$$

Eg 3 $C_2H_5\ddot{O}H > H_2\ddot{O}$ Nucleo.... & Bas...

Eg 4 $Ph-\ddot{O}H < H_2\ddot{O}:$ " " " "

Eg 5 $CH_3\overset{\ominus}{N}H > \overset{\ominus}{N}H_2 >$  $>$  " "

Eg $CH_3-NH_2 > NH_3 >$  $>$  NH^{--} & Bas...

Eg  NH^- & Basicity

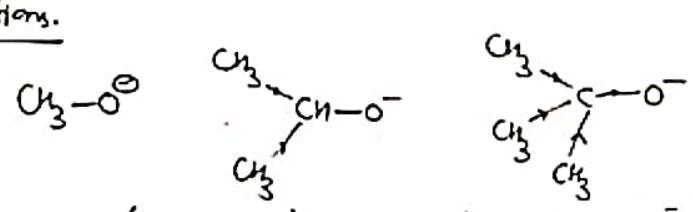
ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 2. When donor atoms are same in different species then order of nucleophilicity and order of basicity are follow same order.

Basic strength $\propto \frac{1}{\text{Stability}}$

exceptions.

① 

, < , < , = Basic Strength

, > , > , = Nucleophilicity (due to steric hindrance)

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 2. When donor atoms are same in diffirent species then order of nucleophilicity and order of basicity are follow same order.

$$\text{Basic strength} \propto \frac{1}{\text{Stability}}$$

exceptions.

② NH_2-NH_2 < NH_3 Basic strength

$NH_2-\overset{\oplus}{N}H_3$ $\quad$ $\overset{\oplus}{N}H_4$

NH_2-NH_2 > NH_3 Nucleophilicity

③ $H-O-O-H$ < H_2O Basi...

.. .. > .. N4C—

④ $HO-NH_2$ < NH_3 Basi---

: > .. Nucl...

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 3. When donor atoms are related to same period.

Left ⟶ Right $\left[\begin{array}{l}\text{Basic strength} \downarrow es \\ \text{Nucleophilicity} \downarrow es\end{array}\right]$ due to increas electronegativity

eg 1 $\ddot{C}H_3^{\ominus} > \ddot{N}H_2^{\ominus} > \ddot{O}H^{\ominus} > \ddot{F}^{\ominus}$ (Nucleophicity and basic strength)

eg 2 $\ddot{N}H_3 > H_2\ddot{O}$ (" " " ")

eg 3 $\ddot{P}H_2^{\ominus} > \ddot{S}H^{\ominus} > \ddot{C}l^{\ominus}$ (" " " ")

eg 4 $\ddot{P}H_3 > H_2S$ " " " "

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 4 When donor atoms are related to same group.

(Top to Bottom)

- Basic Strength ↓es
- Nucleophicity ↑es (in polar protic solvent eg. H_2O) due to solvation.
- Nucleophicity ↓es (in polar aprotic solvent eg. DMSO)

Eg: $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$ Basic Strength.

$F^{-} < Cl^{-} < Br^{-} < I^{\ominus}$ (Nucl---- in polar protic solvent eg H_2O)

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 4 When donor atoms are related to same group.

eg 2 $HO^{\ominus} > SH^{\ominus}$ Basic Strength.

$HO^{-} < SH^{\ominus}$, Nucleophilicity (in polar protic solvent)

$HO^{\ominus} > SH^{\ominus}$, , , (PAS --)

Subscribe

ORGANIC REACTION MECHANISM-I (ORM-I)

Rules for comparison:-

Rule 5 For same species in different solvents.

$Cl^{\ominus}_{aq.} < Cl^{\ominus}_{DMSO}$ (Nucleophilicity)

Solvation — No solvation.

ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- Electrofugality / Nucleofugality ✓

This term is useful for substitution reactions.

$$R-L + G \xrightarrow{\text{Substitution Rxn}\ (S_N \text{ or } S_E)} R-G + L$$

Substrate — Attacking reagent — Leaving group

Imp = S_N = Nucleophilic Substitution Rxn

$$R-L + \overset{\ominus}{Nu} \longrightarrow R-Nu + \overset{\ominus}{L}$$

leaving group (leave with bonding e⁻(s)) ⇒ Nucleofugality

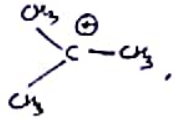
⇒ S_E = Electrophilic Substitution Rxn

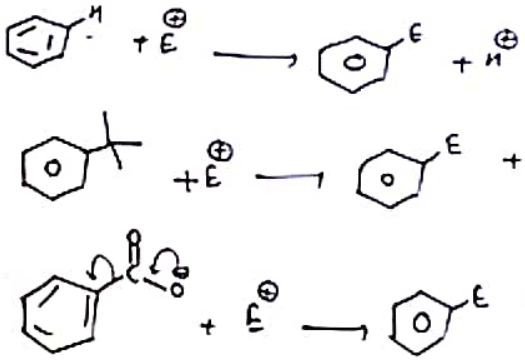
$$R-L + E^{\oplus} \longrightarrow R-E + \overset{\oplus}{L}$$

leaving group (leave without bonding e⁻(s)) ⇒ Electrofugality

ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- Electrofugality / Nucleofugality ✓

→ Examples of Electrofugal group :- $H^{\oplus}$,  , CO_2 SO_3 etc



ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- Electrofugality / Nucleofugality ✓

⇒ Examples of Nucleofugal group → F^-, Cl^-, Br^-, I^-, $R-SO_3^-$, $R-COO^-$, N_2 ✓ H_2O etc.

Leaving group ability ∝ stability of leaving group

$\bar{F}$ → high Charge density

Eg.1

$$R-Cl \xrightarrow[R_1]{\overset{\ominus}{OH}} R-OH + \overset{\ominus}{Cl}$$

$$R-Br \xrightarrow[R_2]{"} " + Br^-$$

$$R-I \xrightarrow[R_3]{"} " + \overset{\ominus}{I}$$

Rate → $R_3 > R_2 > R_1$

l.g. ability → $\overset{\ominus}{I} > Br^- > \overset{\ominus}{Cl}$

$\dot{I}$ — more stable

ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- → Electrofugality / → Nucleofugality ✓

⇒ Examples of leaving group → F^-, Cl^-, Br^-, I^-, $R-SO_3^-$, $R-COO^-$, N_2 ✓ H_2O etc.

CORRECTION- LEAVING ABILITY DIRECTLY PROPORTIONAL TO STABILITY OF LEAVING GROUP

high charge density

Eg.: R– → R–OH + $Cl^\ominus$ / " + Br^- / " + $I^\ominus$ } i.g.

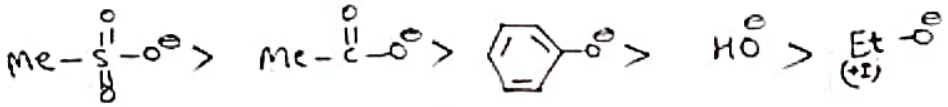
Rate = $R_3 > R_2 > R_1$

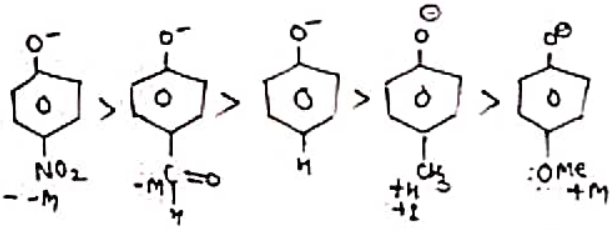


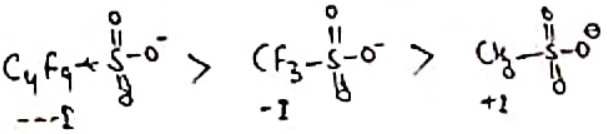
ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- → Electrofugality, → Nucleofugality ✓

Q Give order of leaving group ability.

① 

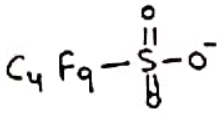
(2) 

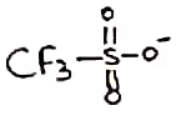
(3) 

ORGANIC REACTION MECHANISM - I ORM-I

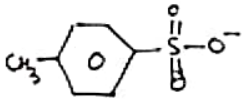
(I.E) Leaving Group Ability :- Electrofugality / Nucleofugality ✓

⇒ Some examples of good leaving groups

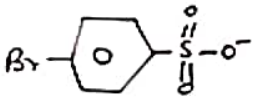




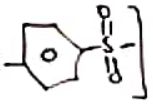
Super l.g.
Triflate (Tf-)



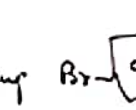
Tosylate (TsO^-)



Brosylate $Bs\text{-}O^-$

Tosyl group []

Ts^-

[Brosyl group ]

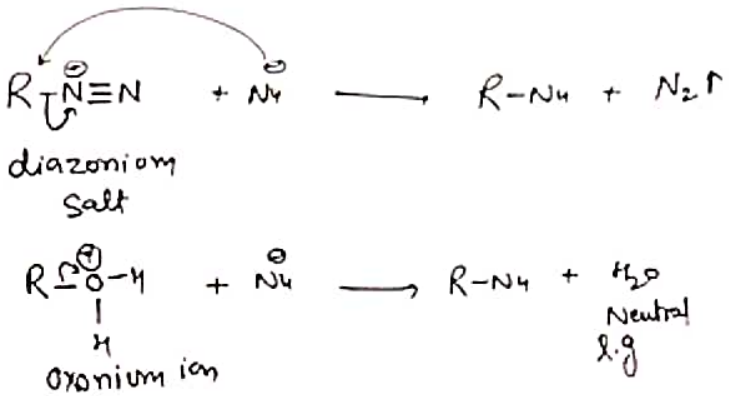
Bs^-

N_2 H_2O

V.V. good l.g good l.g

ORGANIC REACTION MECHANISM - I ORM-I

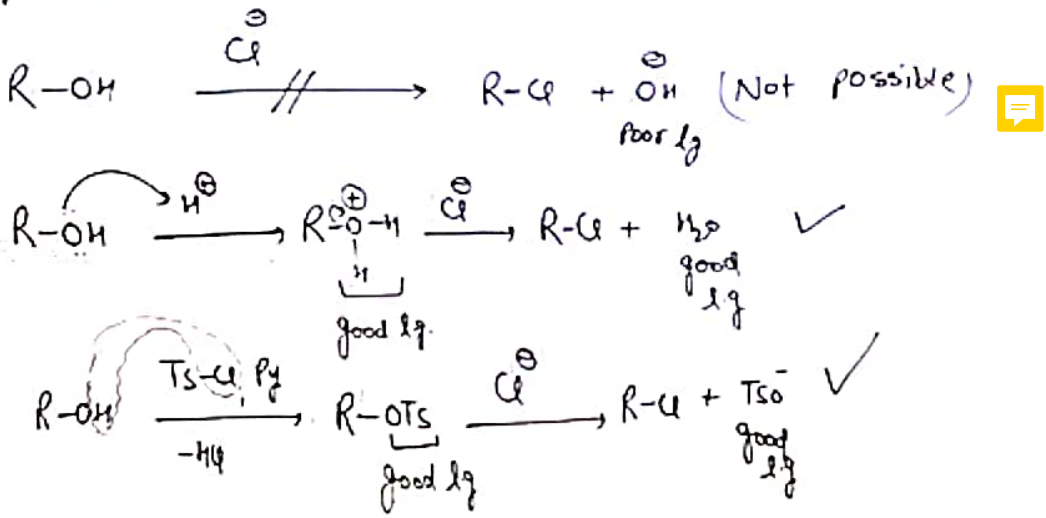
(I.E) Leaving Group Ability :- → Electrofugality / → Nucleofugality ✓



ORGANIC REACTION MECHANISM - I ORM-I

(I.E) Leaving Group Ability :- Electrofugality / Nucleofugality ✓

application



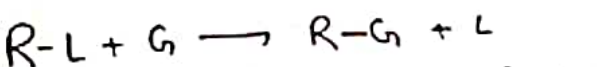
ORGANIC REACTION MECHANISM - I ORM-I

(I-F) Types of reactions :-

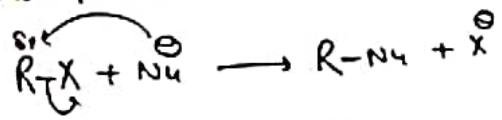
(A) Acid - base Rxn



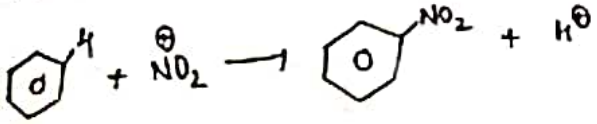
(B) Substitution Reaction = Displacement reaction



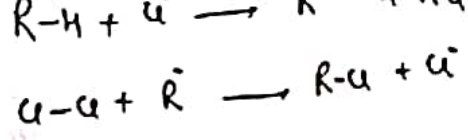
B1 Nucleophilic Substitution Reaction (S_N)



B2 Electrophilic Sub Rxn (S_E)



B3 Free radical sub. Rxn

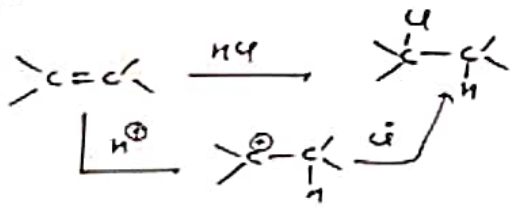


ORGANIC REACTION MECHANISM - I ORM-I

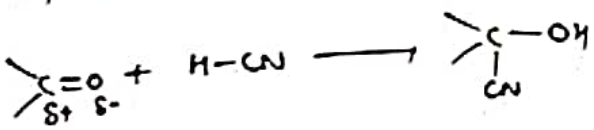
Types of reactions :-

(C) Addition Rxn C=C , C≡C >C=O

C.1 Electrophilic Addition :-



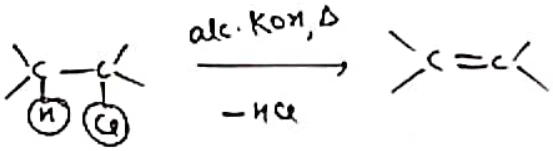
C.2 Nucleophilic Addition :-



C.3 - Free radical Addition :-

(I-F) Types of reactions :-

(D) Elimination Reaction :-

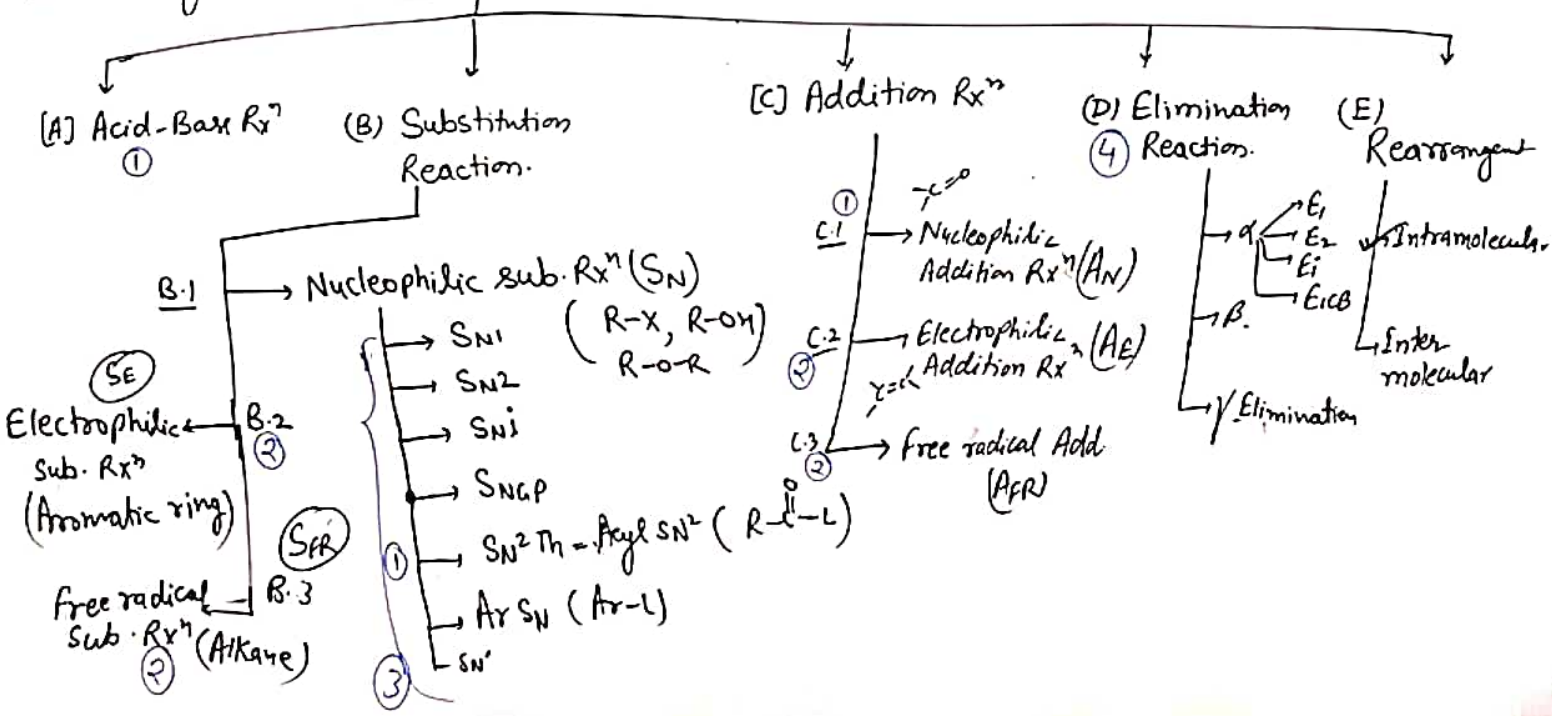


(E) Rearrangent Reaction :-



ORGANIC REACTION MECHANISM - I ORM-I

(I-F) Types of reactions :-



ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions :-

$$\underset{\text{Acid}}{A-H} + \underset{\text{Base}}{B^{\ominus}} \longrightarrow \underset{\text{Base}}{A^{\ominus}} + \underset{\text{acid}}{B-H}$$

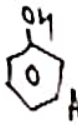
(Condition : $A-H > B-H$ acidic strength; $B^{\ominus} > A^{\ominus}$ Basic strength)

⇒ Favourable condition for acid-base Rx^{n}

Acid and base of reactant side should be stronger than acid & base of product side.

Eg. CH_3-COOH + $NaOH$ ⇌ $CH_3-COO^{\ominus}Na^{\oplus}$ + H_2O

Acid (base) (Base) (Acid)

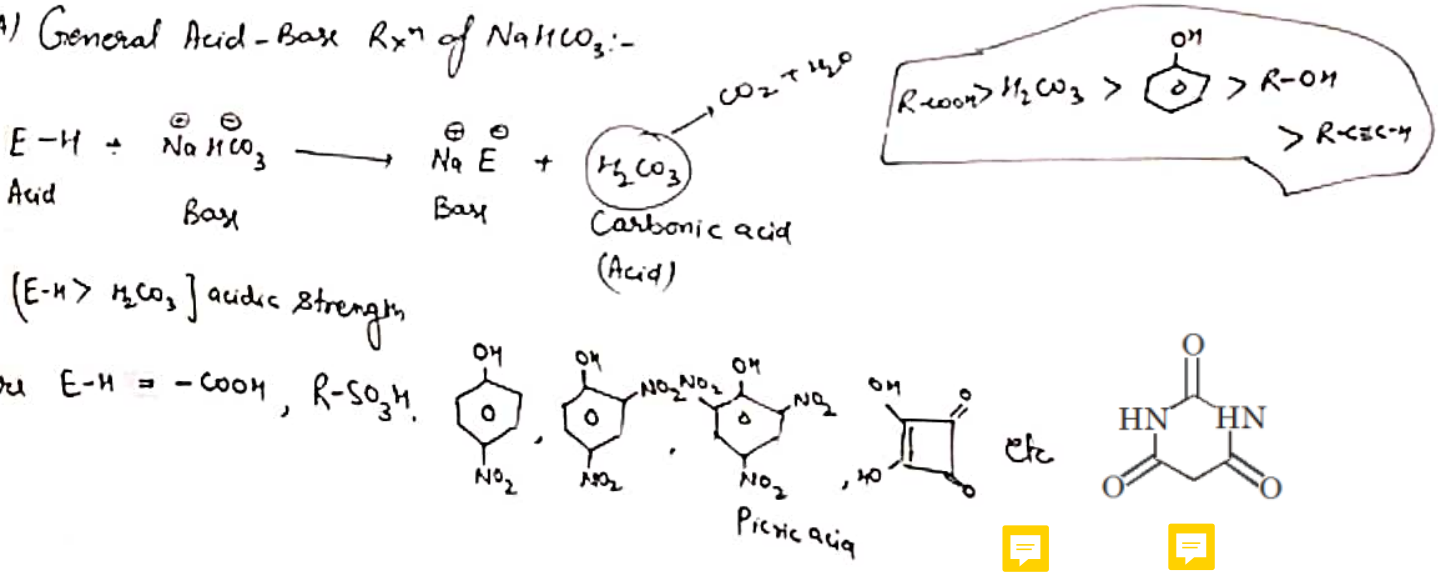
Eg CH_3-COOH +  ⇌ CH_3COONa + 

Acid (Base) (Base) (Acid)

ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions :-

(II-A) General Acid-Base Rxn of $NaHCO_3$:-



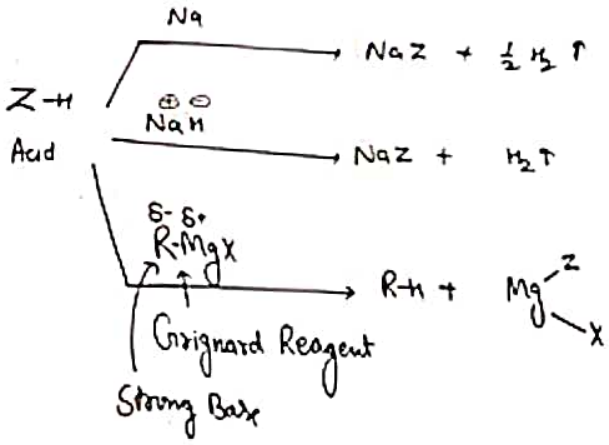
(i) increasing acidity :

CH_3CH_2OH < CH_3CH_2SH < CH_3CO_2H < CCl_3CO_2H Resonance stabilisation in carboxylates makes carboxylic acids more acidic than alcohols or thiols. Electron withdrawing groups, here -Cl, increase the acidity due to further stabilisation of the carboxylate.
Thiols are more acidic than alcohols due to weaker X-H bond and the ability of S to accommodate extra electrons (size).

ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

(II.B) Acid-Base Reaction of Active Hydrogens:-



Where Z-H =

$-\overset{\delta-}{O}-\overset{\delta+}{H}$ { H_2O, D_2O, $R-\overset{O}{\overset{\|}{C}}-OH$, $R-\overset{O}{\overset{\|}{\underset{O}{S}}}-OH$, $R-OH$, $Ar-OH$, $>C=N-OH$ etc

$-\underset{|}{N}-H$ { NH_3, $R-NH_2$, R_2N-H, $R-\overset{O}{\overset{\|}{C}}-NH_2$ etc, $Ar-NH_2$)

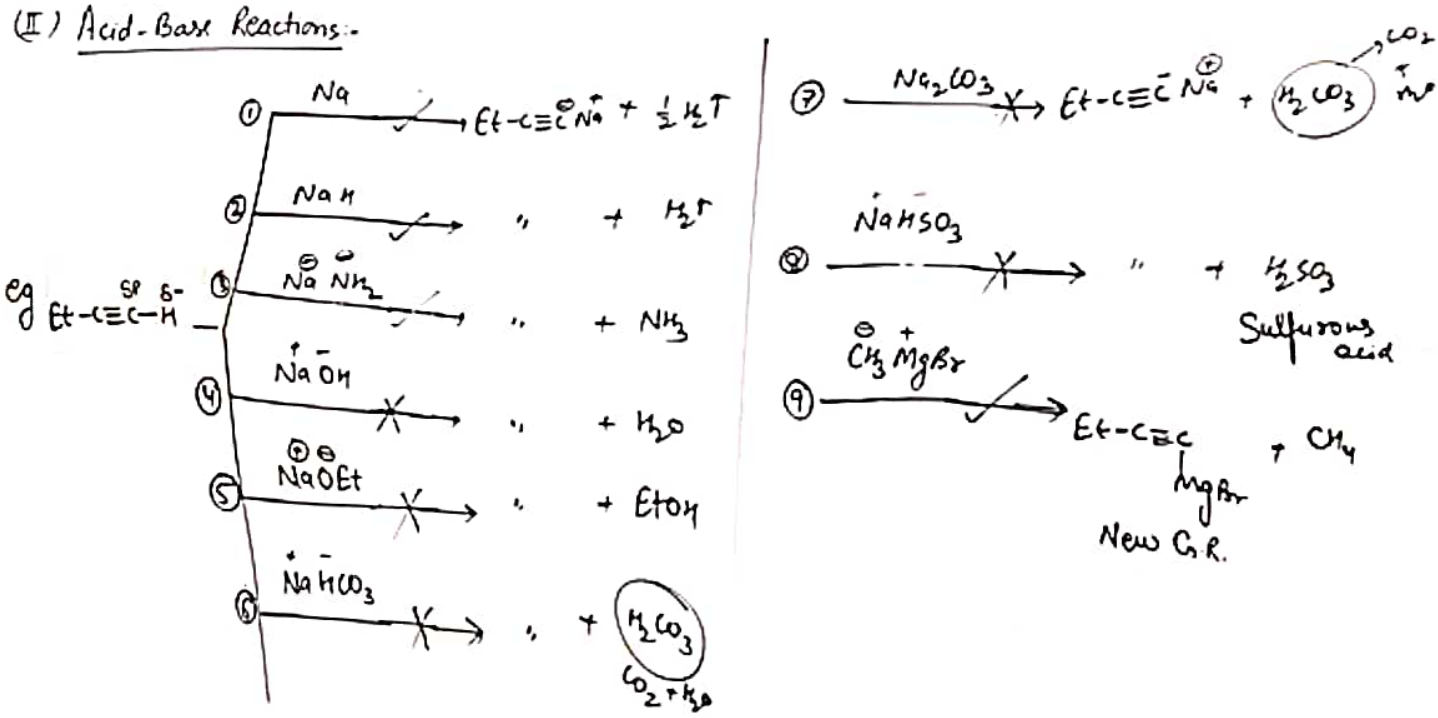
$X-H$ (H-F, HCl, HBr, HI)

$-SH$ (H_2S, R-SH, Ar-SH)

$-C\equiv C-H$ (sp), $-CH_2-$(-M), (-M) AMG, etc.

ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

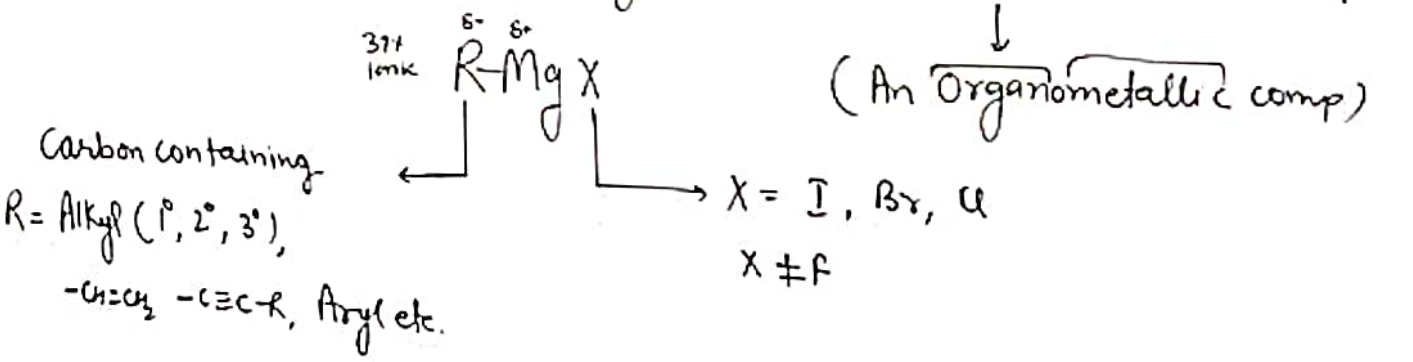


ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

(II.C) Acid-Base Reactions of Grignard Reagent:-

Introduction of Grignard Reagent



ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

- Formation of $RMgX$

$$R-X + Mg \xrightarrow[\text{solvent}]{\text{dry ether}} R-MgX + \text{Heat}$$

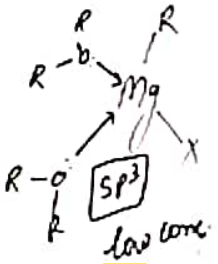
(R-X: ⇓; Mg: excess)

⇒ Reactivity of R-X with Mg → R-I > R-Br > R-Cl >>> R-F

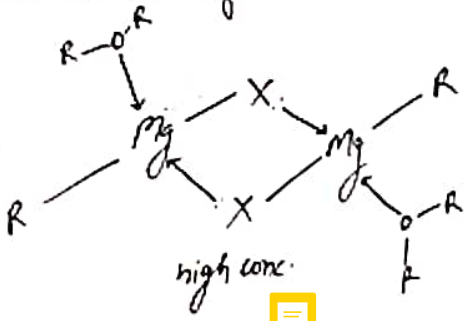
No R_x^n with Mg

⇒ Use of Ether:- ① Solvent

② Complet octet of Mg



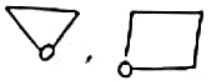
or



ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

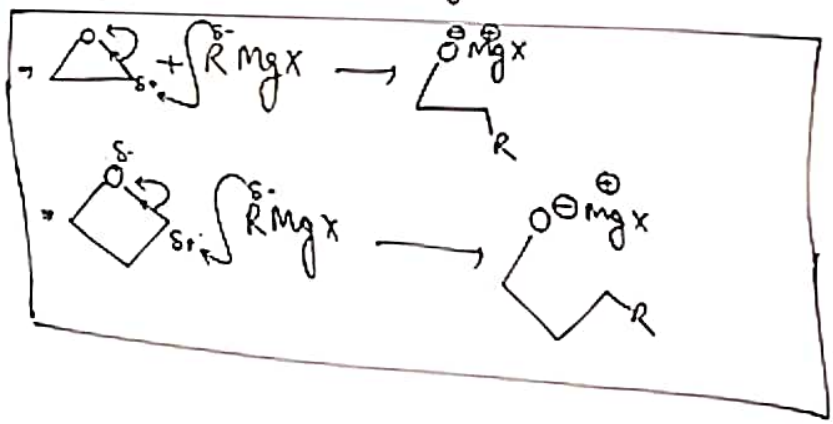
Note



, Can not be used as solvent for Grignard Reagent

(highly reactive due high ring Strain)

→ Can give reaction with RMgX



ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

⇒ Reactivity of RMgX w.r.t X ⇒ R-MgI > RMgBr > RMgCl

* Reactivity " " w.r.t R ⇒ $C_2H_5^{\delta-}MgX$ (sp^3) > $CH_2=CH^{\delta-}-Mg^{\delta+}X$ (sp^2) > $R-C\equiv C^{\delta-}-Mg^{\delta+}X$ (sp)

Stronger $N_u^{\ominus}$ — weaker $N_u^{\ominus}$

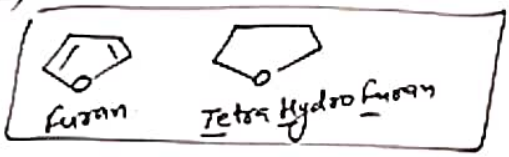
⇒ Reactivity of various functional groups with RMgX

Acidic Hydrogen > $R-\overset{O}{\overset{\|}{C}}-H$ > $R-\overset{O}{\overset{\|}{C}}-R$ ≥ $R-\overset{O}{\overset{\|}{C}}-Cl$ > $R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$ > $R-\overset{O}{\overset{\|}{C}}-OR$ > R-X

Acid-Base Rx' — Nucleophilic Addition — S_N2 Th — S_N2

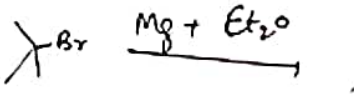
(R-X) + Mg → RMgX

ORGANIC REACTION MECHANISM - I (ORM-I)



(II) Acid-Base Reactions:-

eg1 $Me-Br \xrightarrow{Mg + Et_2O} MeMgBr \xrightarrow{H_2O} CH_4 + Mg(Br)(OH)$

eg2 

eg3 $CH_2=CH-Cl \xrightarrow[(THF)]{Mg + \text{furan}} CH_2=CH-MgCl \xrightarrow{CH_3-\overset{O}{\overset{\|}{C}}-OH} CH_2=CH_2 + Mg(Cl)(O-\overset{O}{\overset{\|}{C}}-CH_3)$

$CH_2=CH-Br \xrightarrow{Mg + Et_2O} CH_2=CH-MgBr$ ✓

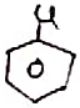
$CH_2=CH-Cl \xrightarrow{Mg + Et_2O} X$

ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

eg 4 $CH_3-C\equiv C-Br + Mg \xrightarrow{THF} CH_3-C\equiv C-MgBr \longrightarrow CH_3-C\equiv C-H$ + 

alkyne — New G.R

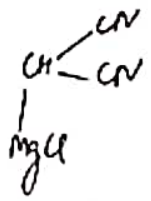
eg 5  $+ Mg \xrightarrow{THF} PhMgCl \xrightarrow{CH_3-C\equiv C-H} Ph-H + CH_3-C\equiv C-MgCl$

Benzene — New G.R

eg 6 $CH_3-O-CH_2-Cl + Mg \xrightarrow{ether} CH_3-O-CH_2MgCl \xrightarrow{CH_3-\overset{O}{\overset{\|}{C}}-NH_2} CH_3-O-CH_3 + CH_3-\overset{O}{\overset{\|}{C}}-NH-MgCl$

ether

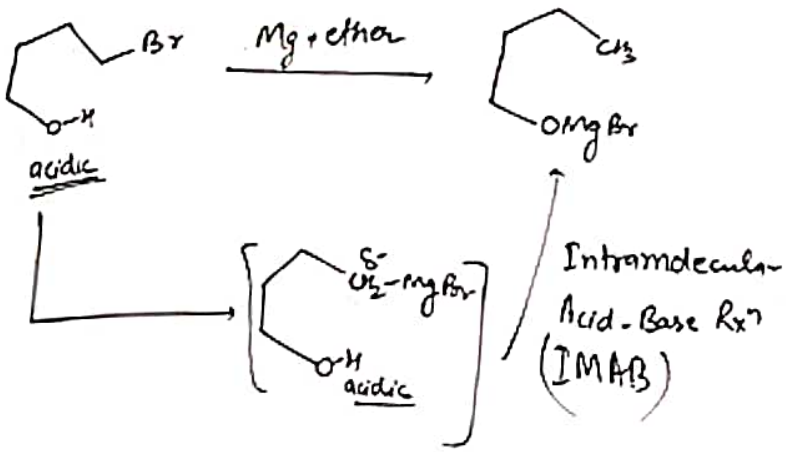
eg 7  $+ Mg \xrightarrow{Et_2O}$ 

$\xrightarrow[AMS]{CH_2=CN(M)}$  + 

ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

Note



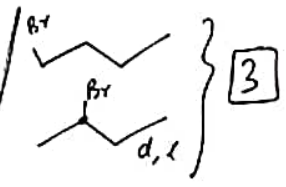
ORGANIC REACTION MECHANISM - I (ORM-I)

(II) Acid-Base Reactions:-

Q. $R-Br \xrightarrow{Mg + ether} RMgX \xrightarrow{H_2O}$ R-H

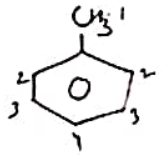
Possible alkyl bromide ?

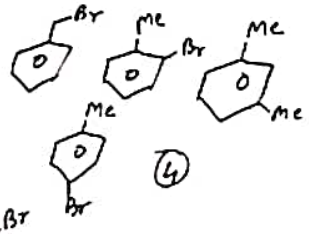
n-Butane



3

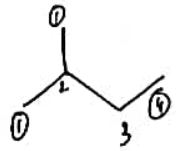
Q. $R-Br$ (Possible Bromide) $\xrightarrow{Mg + ether} RMgBr \xrightarrow{H_2O}$

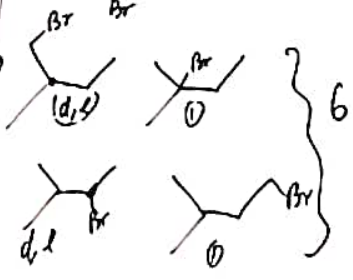




4

Q. $R-Br$? $\xrightarrow{"} R-MgBr \xrightarrow{H_2O}$



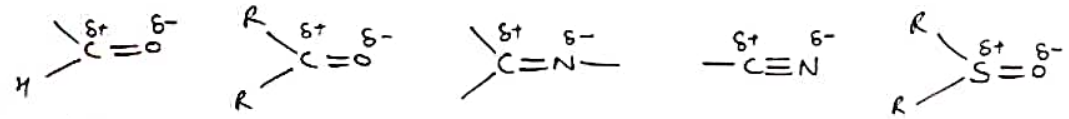


6

ORGANIC REACTION MECHANISM - I (ORM-I)

(III) NUCLEOPHILIC ADDITION REACTIONS ($A_N R_x^n$)

This reaction is given by polar multiple bond containing comp.



Aldehyde Ketone Imine Cyanide

ORGANIC REACTION MECHANISM - I (ORM-I)

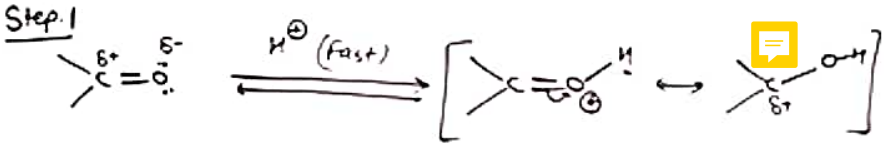
(III) NUCLEOPHILIC ADDITION REACTIONS (A_N Rxn)

Mechanisms for A_N on carbonyl comp.

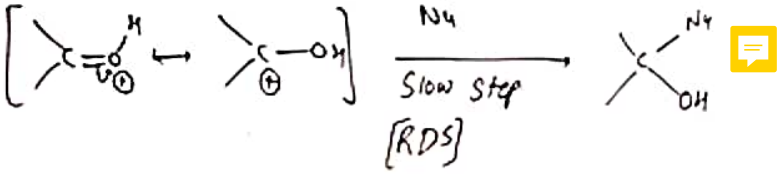
RDS = Rate Determing Step

(a) In acidic medium:-

Step-1



Step-2



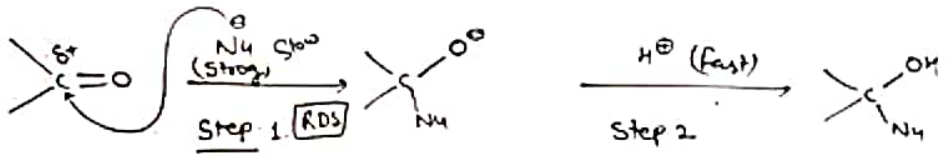
ORGANIC REACTION MECHANISM - I (ORM-I)

(III) NUCLEOPHILIC ADDITION REACTIONS (A_N Rx^n)

Mechanisms for A_N on carbonyl comp.

RDS = Rate Determing Step

(b) In basic medium (In absence of acid / in presence of strong $Nu^{\ominus}$)



ORGANIC REACTION MECHANISM - I (ORM-I)

(III) NUCLEOPHILIC ADDITION REACTIONS (A_N Rx^n)

General introduction of Chemical Kinetics



① (i) Simple Rx^n (Elementry Rx^n)

$$A + 2B \longrightarrow AB_2$$

Rate $\propto [A]^1[B]^2$ ∴ 3rd order Rx^n

③

$$2B \xrightarrow{RDS} B_2$$
$$B_2 + A \longrightarrow AB_2$$

Rate $\propto [B]^2$

② (ii) Complex Rx^n

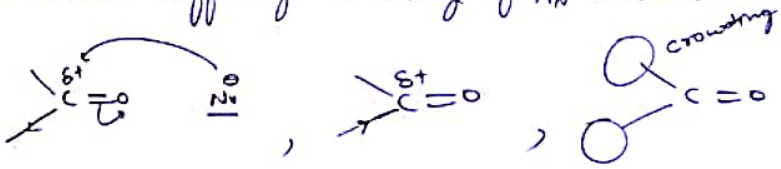
$$A + B \xrightarrow{Slow} AB$$
$$AB + B \xrightarrow{Fast} AB_2$$
$$A + 2B \longrightarrow AB_2$$

(Rate of Reaction = Rate of RDS) $\propto [A]^1[B]^1$

ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

⇒ Factors affecting reactivity of A_N reactions.



(1) Reactivity of A_N reaction ∝ Nucleophilicity of ($Nu^{\ominus}$)

(2) " " " " " ∝ electrophilicity of carbonyl group.
or (magnitude of +ve charge on carbon of carbonyl group.

(3) " " " " " ; ↑ in presence of e^- withdrawing groups (−M, −I)

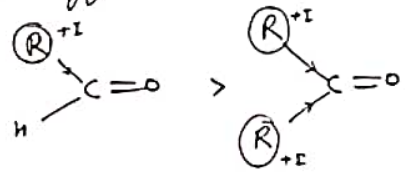
(4) " " " " ↓ " " " e^- donating groups (+M +H +I)

(5) " " " " " ↓ " " " bulky groups (Steric hindrance)

ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

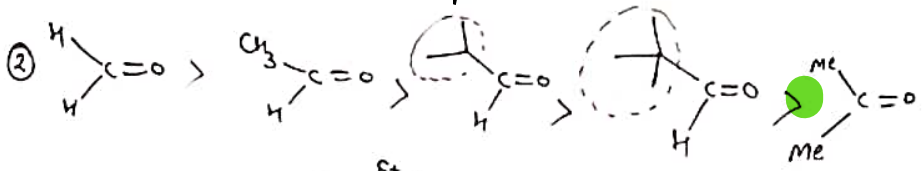
Q Identify order of reactivity for A_N Rxn.

① 

Aldehyde > Ketones

isobutyraldehyde

Pivaldehyde/
trimethyl acetaldehye

② 

(3) $CH_3-CH=O < CH_2Cl-CH=O < CHCl_2-CH=O < CCl_3-CHO$

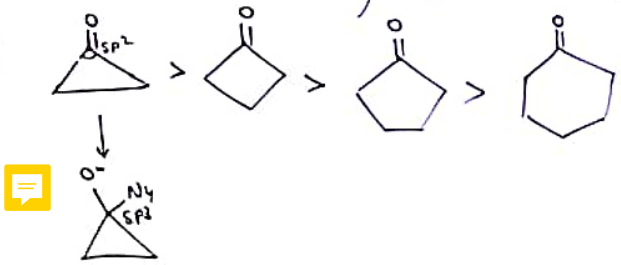
Trichloroacetaldehyde
(Chloral)

ORGANIC REACTION MECHANISM -I (ORM-I)

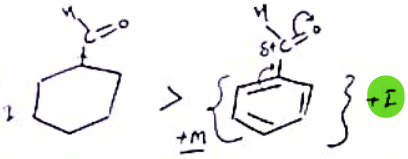
(III) Nucleophilic Addition Reactions (A_N)

Q Identify order of reactivity for A_N Rx^n

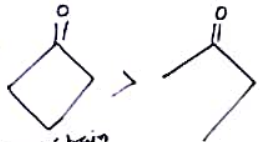
(4) $CH_3 \xrightarrow{+I} \overset{\delta+}{C}H=O$ < Glyoxal < most reactive

(5) 

reactivity (due to ring Strain)

(6) 

Aliphatic carbonyl comp. > Aromatic carbonyl comp.

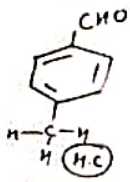
(7) 

Cyclic Ketone > Acyclic Ketone

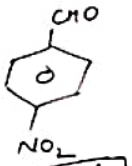
ORGANIC REACTION MECHANISM -I (ORM-I)

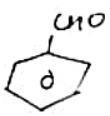
(III) Nucleophilic Addition Reactions (A_N)

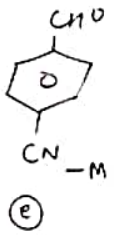
Q. Identify order of reactivity for A_N Rx^n

(a) 

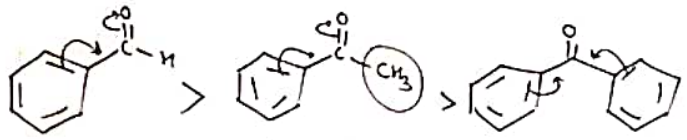
(b) 

(c) 

(d) 

(e) 

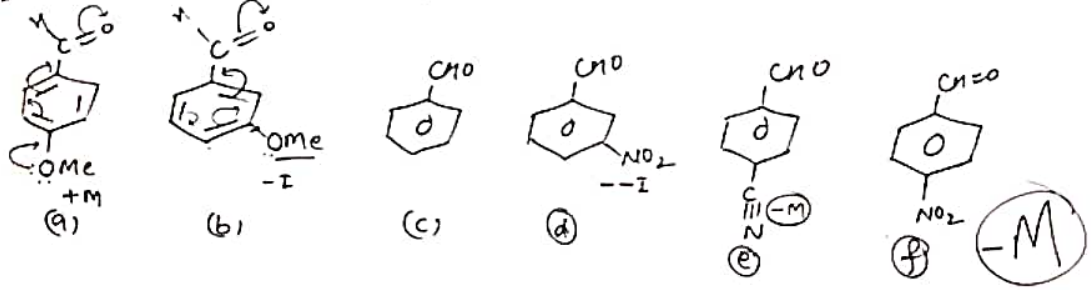
reactivity = c > e > d > a > b

(a) 

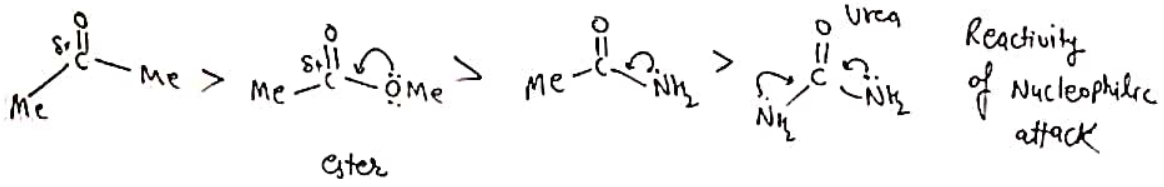
ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

Q Identify order of reactivity for A_N Rxn

→ 

f > e > d > b > c > a

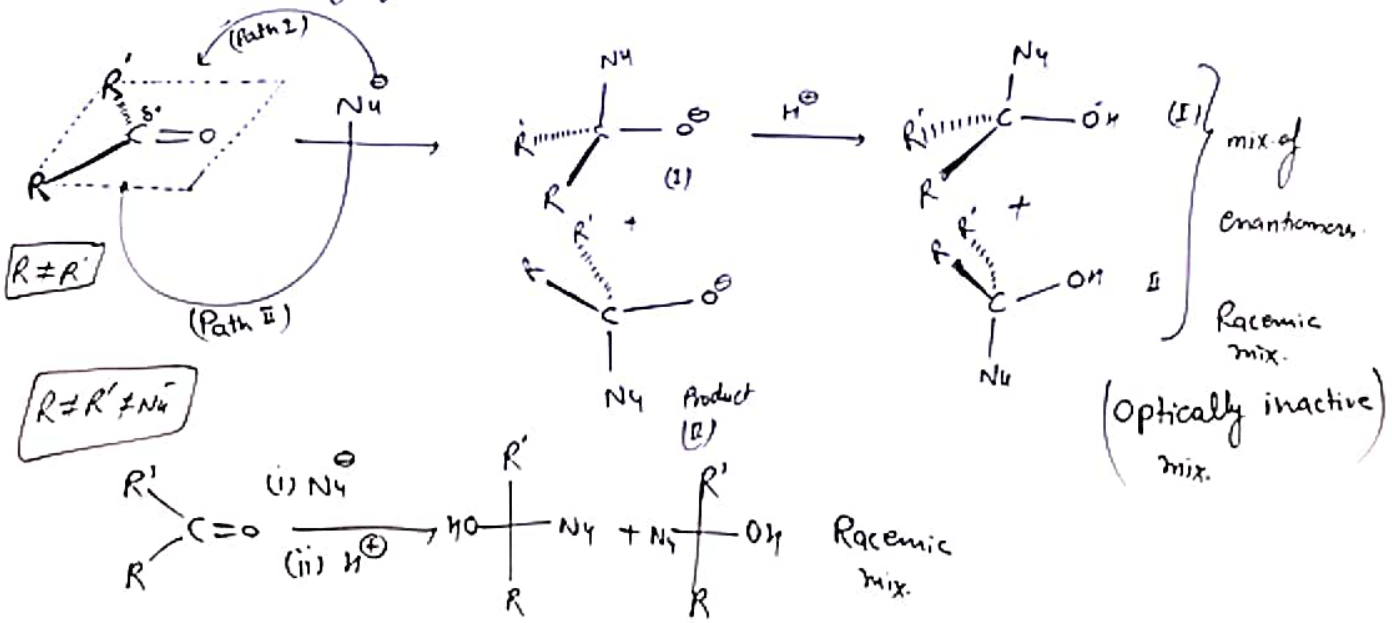
→ 

Reactivity of Nucleophilic attack

ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

Stereochemistry of Nucleophilic addition Rx"

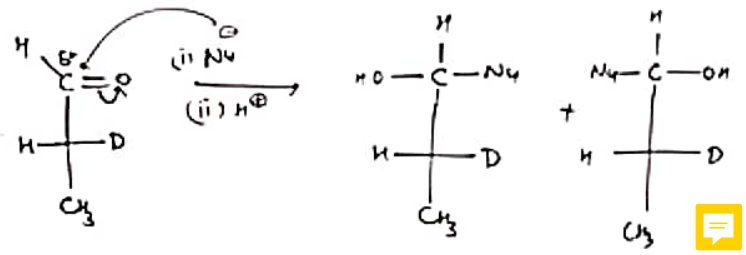


ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

Stereochemistry of Nucleophilic addition Rx"

Case I

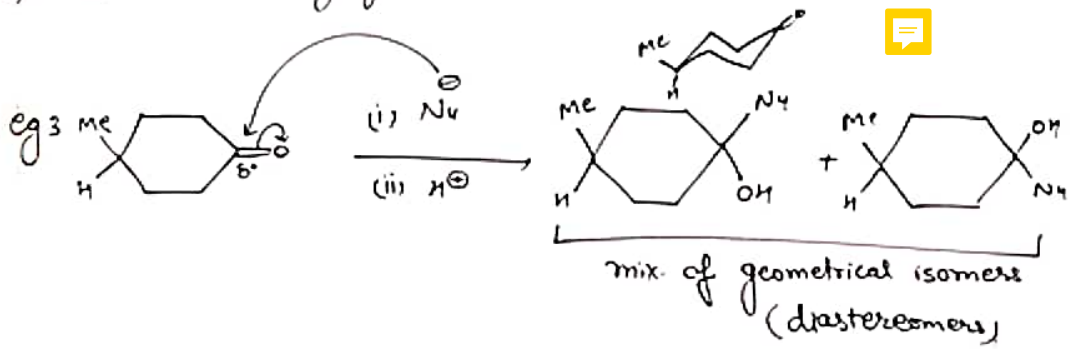


mixture of diastereomers.

ORGANIC REACTION MECHANISM -I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

Stereochemistry of Nucleophilic addition Rx"



mix. of geometrical isomers (diastereomers)

ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(II A) General N_u Addition Reaction of aldehyde, Ketones

(1) Addition of H_2O

(2) " " R-OH

(3) " " HCN

(4) " " $NaHSO_3$

(5) " " $R-C\equiv\overset{\ominus}{C}\,\overset{\oplus}{Na}$

(6) " " " $RMgX$ & RLi

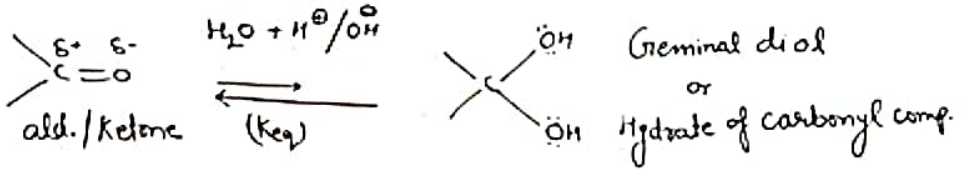
(7) " " " $LiAlH_4$ & $NaBH_4$

ORGANIC REACTION MECHANISM-I (ORM-I)

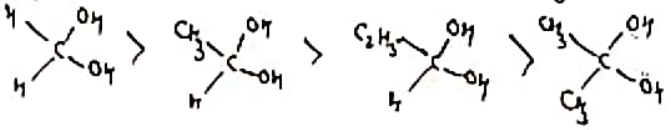
(III) Nucleophilic Addition Reactions (A_N)

(II A) General N_u Addition Reaction of aldehyde, Ketones:

(1) Addition of water:-



Note:- Generally gem. diols are unstable due to lp..lp repulsion of oxygens of -OH.

- Stability of hydrates of carbonyl comp. $\propto \frac{1}{\text{Steric hindrance of carbonyl group}}$
- Stability of of hydrates :- 

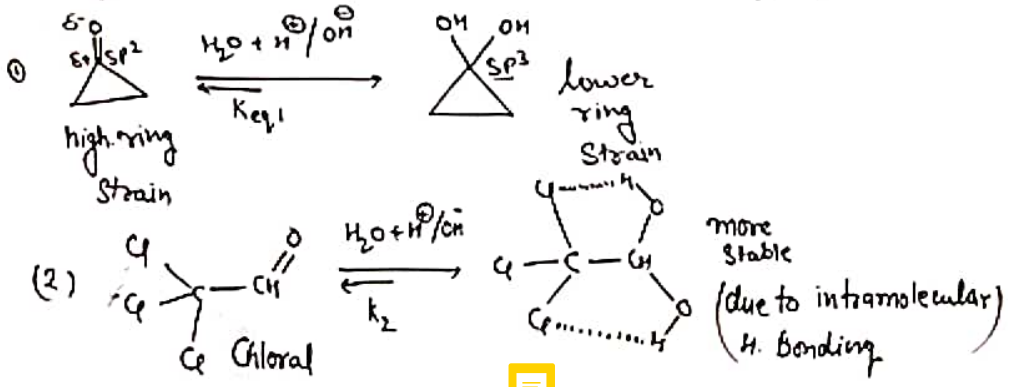
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General N_u Addition Reaction of aldehyde, Ketones:

(1) Addition of water:-

- Following hydrates of carbonyl comp are highly stable



ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

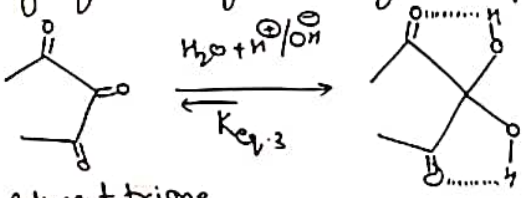
(III A) General Nu. Addition Reaction of aldehyde, Ketones:-

F—H F⁻
O—H :O<
N—H N<
Cl⁻

(1) Addition of water:-

⇒ Following hydrates of carbonyl comp. are highly stable

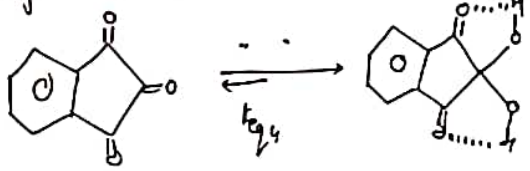
(3)



highly stable due to intramole— H bonding.

Adjacent trione

(4)



Keq.4 > Keq.3 > Keq.2 > Keq.1

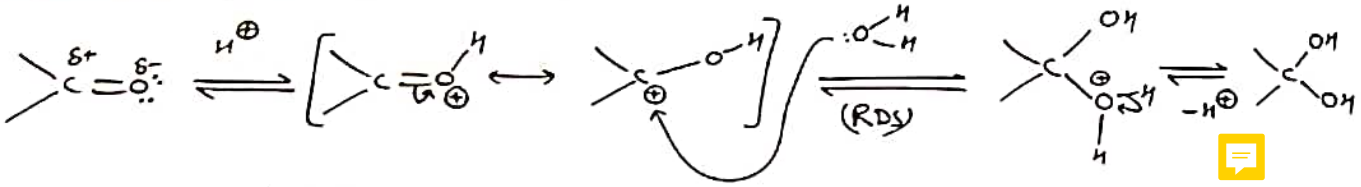
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

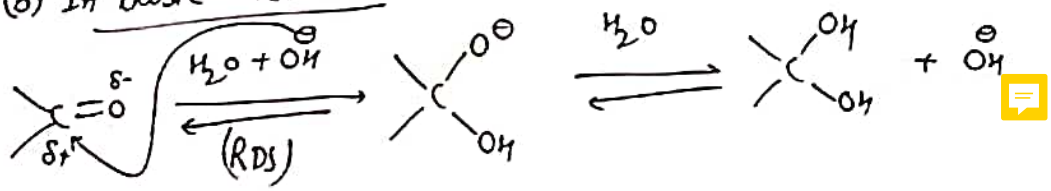
(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(1) Addition of water:-

Mechanism (a) In acidic medium.



(b) In basic medium:-



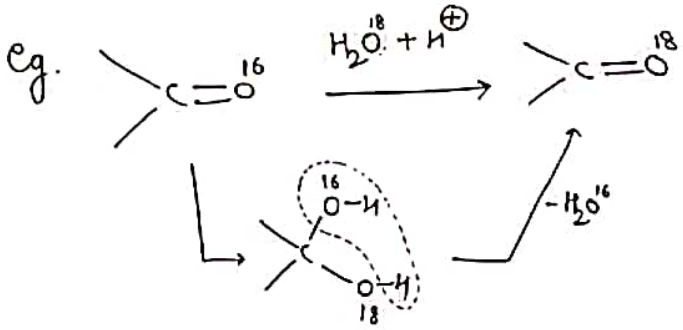
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(1) Addition of water:-

Mechanism (a) In acidic medium

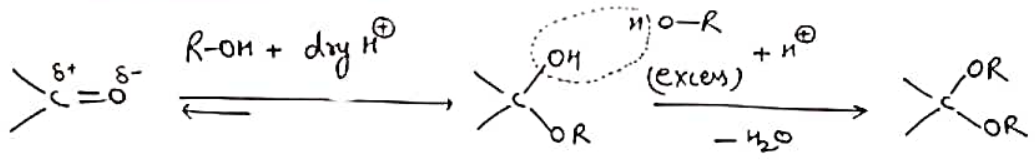


ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(2) Addition of alcohols :- R-O-H



Hemi acetal (ald. reactant)
or
Hemi ketal (when ketone reactant)

Acetal (when reactant aldehyde)
or
Ketal (" " ketone)

Note :-
- {Hemi acetal & Hemi ketal} → Hydrolysis possible in dil. acid or in dil. base
- {Acetal & ketal} → Hydrolysis possible in dil. acidic medium not in basic medium.

- Dry HCl is used to avoid hydrolysis of products.

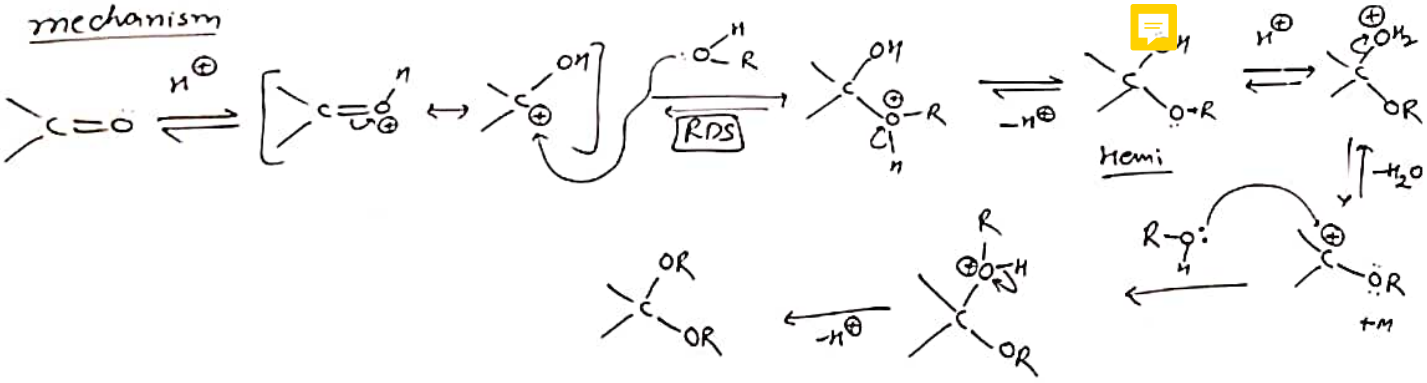
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(2) Addition of alcohols :- R-O-H

mechanism



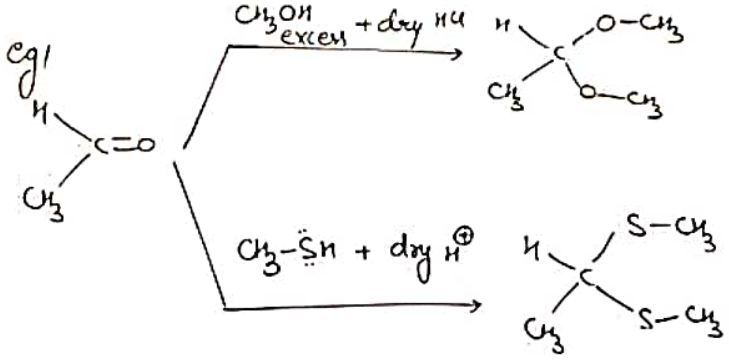
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

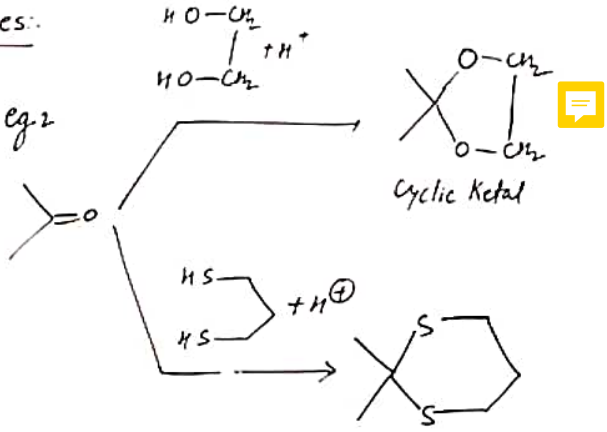
(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(2) Addition of alcohols :- R-O-H

eg1



eg.2



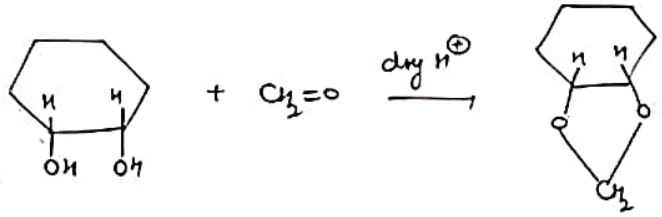
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

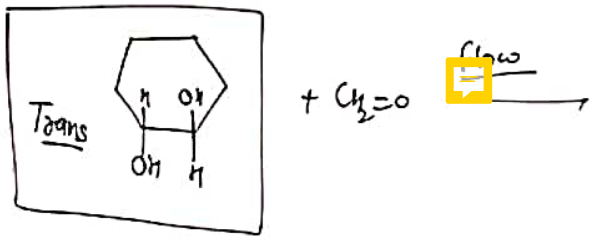
(2) Addition of alcohols :-

Cg 3



Cis

Trans



ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

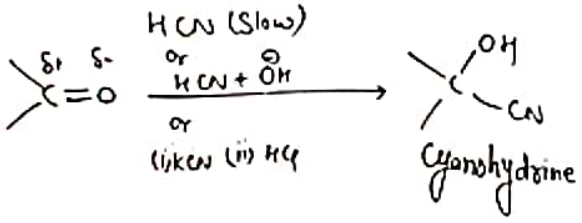
(3) Addition of HCN:-

Reagent ① $HCN \rightleftharpoons H^{\oplus} + \overset{\ominus}{CN}$ (Slow)

weak base — low conc.

✓ (2) $HCN + \overset{\ominus}{OH} \rightleftharpoons H_2O + \overset{\ominus}{CN}$ higher conc

✓ (3) (i) KCN (ii) HCl



Cyanohydrine

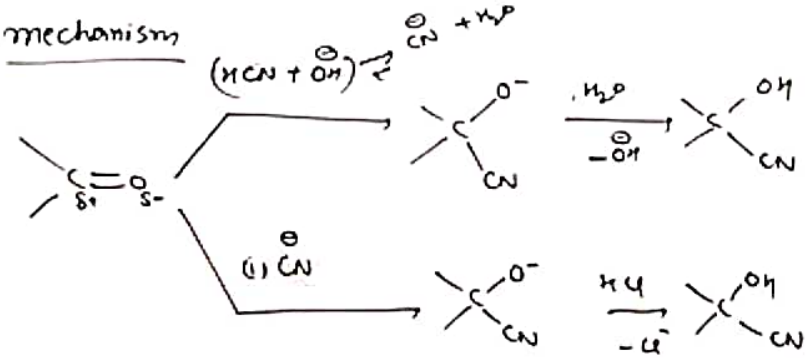
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(3) Addition of HCN:-

mechanism



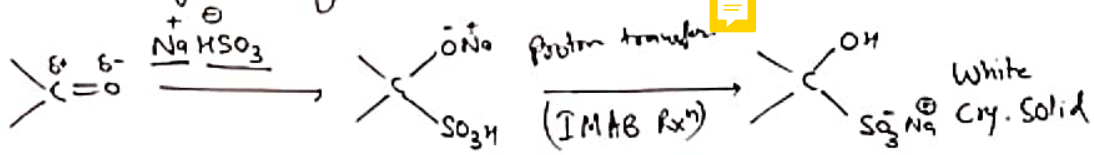
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

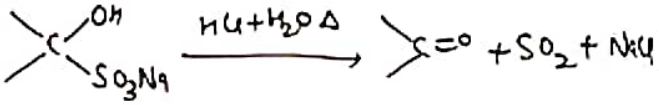
(1) Addition of $NaHSO_3$ (Sod bisulfhite)

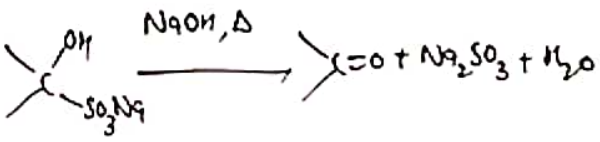
→ Test of aldehydes and small Ketones



⇒ We can form aldehyde and Ketones with product of this reaction (this salt)

In presence of dil. acid or dil. base on high temp.





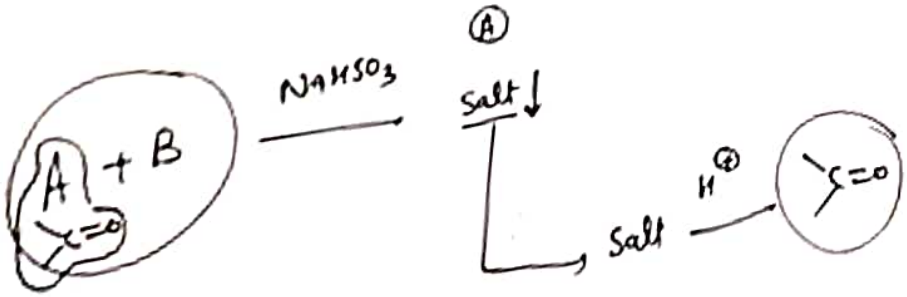
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General N_u Addition Reaction of aldehyde, Ketones

(1) Addition of $NaHSO_3$ (Sod. bisulfhite)

- This method is used for isolation of carbonyl comp.



ORGANIC REACTION MECHANISM-I (ORM-I)

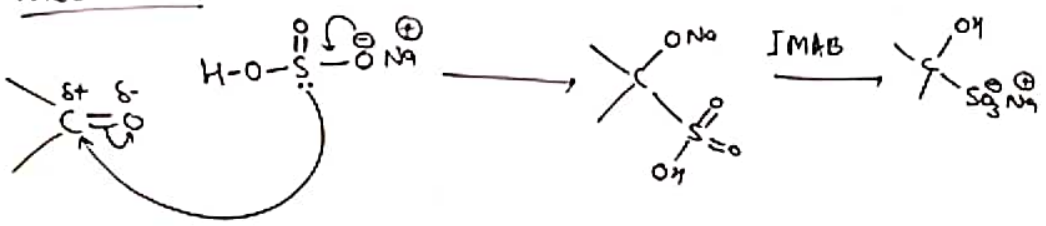
(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

(1) Addition of $NaHSO_3$ (Sod. bisulfhite)

- This method is used for isolation of carbonyl comp.

mechanism

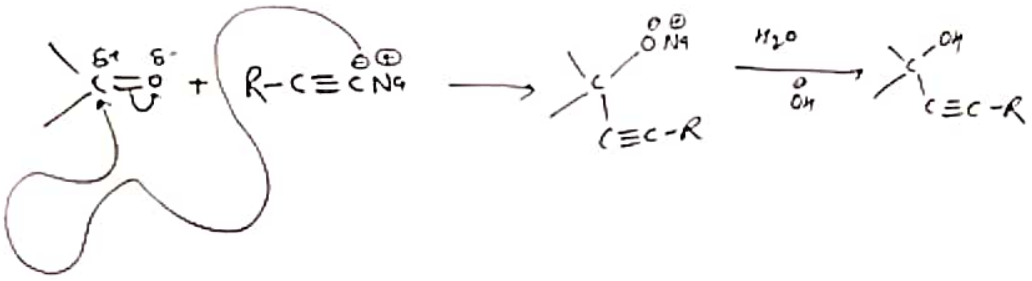


ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones

(5) Addition of alkylides ($R-C\equiv C^{\ominus}\,{}^{\oplus}Na$)



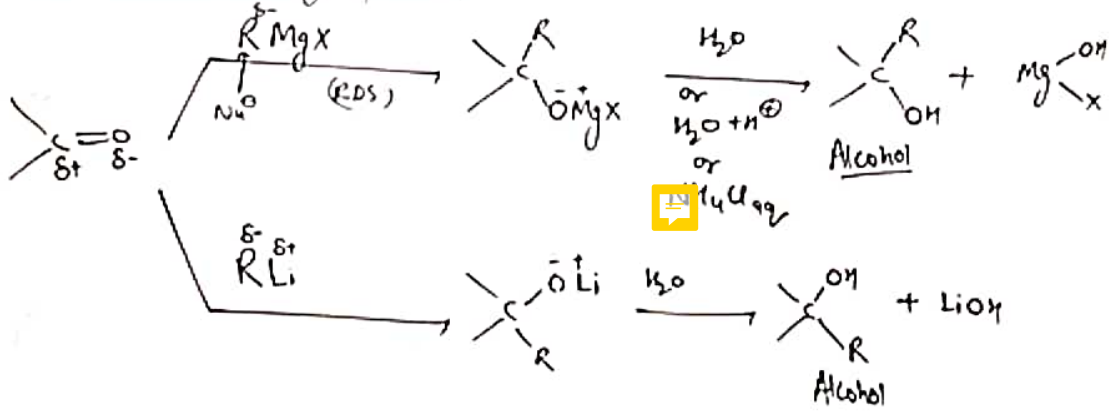
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones

$$R\text{-}X + 2Li \xrightarrow{\text{ether}} RLi + LiX$$

(6) Addition of RMgX / RLi



ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

(III A) General Nu. Addition Reaction of aldehyde, Ketones:

$$R\text{-}X + 2Li \xrightarrow{\text{ether}} RLi + LiX$$

(1) Addition of $LiAlH_4$ / $NaBH_4$ ether solvent

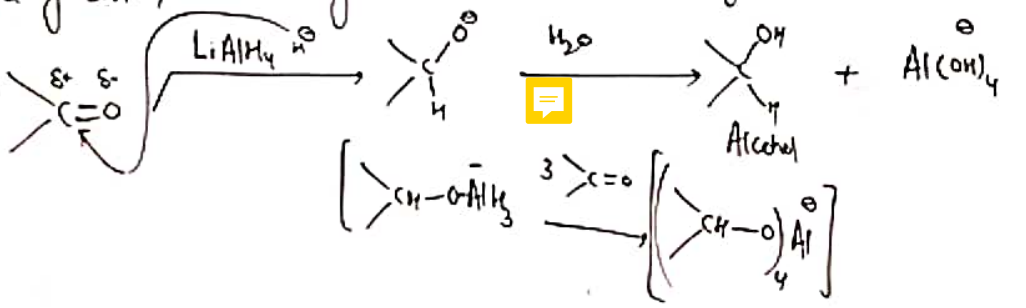
$H^{\ominus}$ $H^{\ominus}$

$$LiH + AlH_3 \longrightarrow LiAlH_4$$

$$NaH + \frac{1}{2}B_2H_6 \longrightarrow NaBH_4$$

$Li(AlH_4^-)$ > $Na^+(BH_4^-)$

Hydride of $LiAlH_4$ is stronger Nucleophile w.r.t. hydride of $NaBH_4$



ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N)

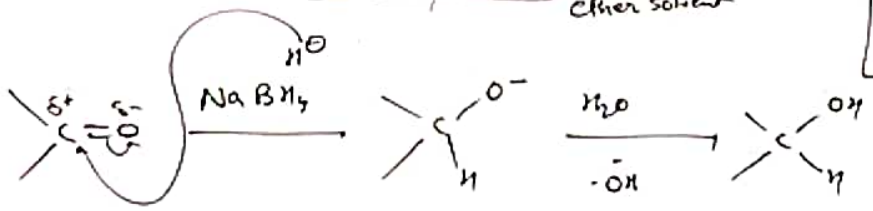
(III A) General Nu. Addition Reaction of aldehyde, Ketones

$$R\text{-}X + 2Li \xrightarrow{\text{ether}} RLi + LiX$$

(1) Addition of $LiAlH_4$ / $NaBH_4$

$$LiH + AlH_3 \longrightarrow LiAlH_4$$

$$NaH + \tfrac{1}{2}B_2H_6 \longrightarrow NaBH_4$$

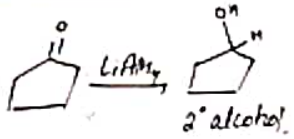


eg $CH_3-\overset{\delta+}{C}H=O$ (Ald.)

$\xrightarrow{\text{(i) } LiAlH_4 \text{ (ii) } H_2O}$ $CH_3-\underset{H}{CH}-OH$ 1° alcohol

$\xrightarrow{\text{(i) } LiAlD_4 \text{ (ii) } H_2O}$ $CH_3-\underset{D}{CH}-OH$ ($\pm$) Racemic mix.

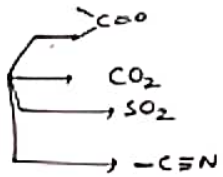
$\xrightarrow{\text{(i) } LiAlH_4 \text{ (ii) } D_2O}$ CH_3-CH_2-OD

eg  2° alcohol

ORGANIC REACTION MECHANISM-I (ORM-I)

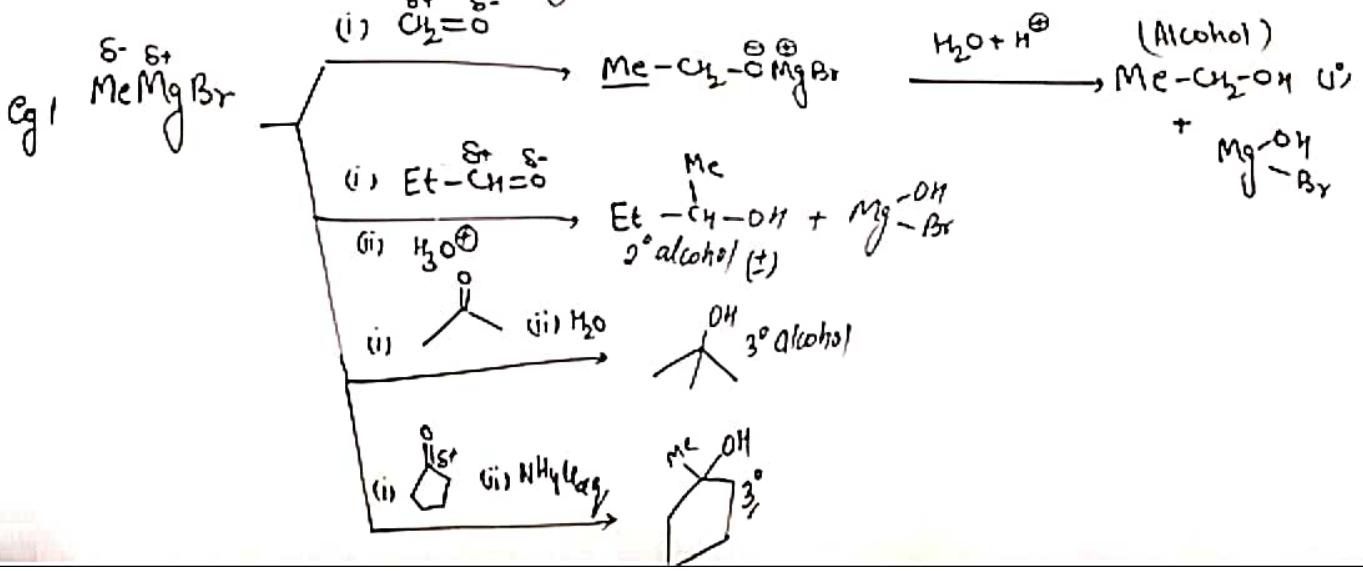
(III) Nucleophilic addition reactions (A_N) :-

(III.B) Nucleophilic addition reactions of $RMgX$:-



and RLi

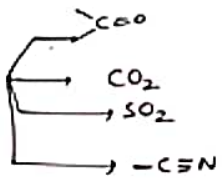
(i) Reactions with carbonyl compound :-



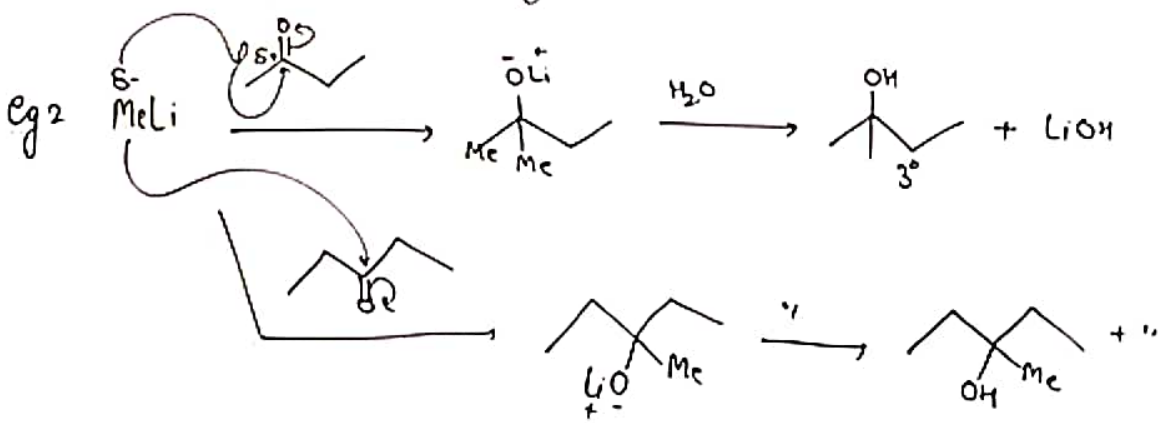
ORGANIC REACTION MECHANISM-I (ORM-I)

(III) Nucleophilic addition reactions (A_N) :-

(III.B) Nucleophilic addition reactions of $RMgX$ and RLi :-



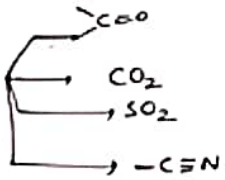
(i) Reactions with carbonyl compound :-



ORGANIC REACTION MECHANISM-I (ORM-I)

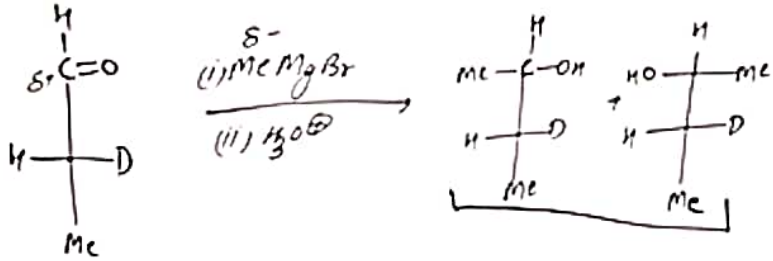
(III) Nucleophilic addition reactions (A_N) :-

(III.B) Nucleophilic addition reactions of $RMgX$ and RLi :-



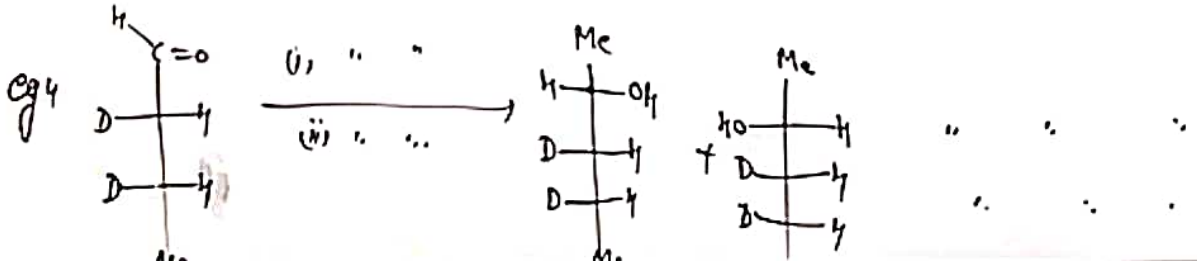
(i) Reactions with carbonyl compound :-

eg 3



mix. of diastereomers (optical active mix.)

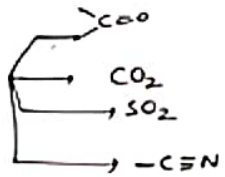
eg 4



ORGANIC REACTION MECHANISM-I (ORM-I)

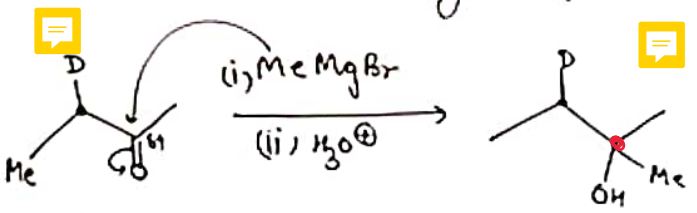
(III) Nucleophilic addition reactions (A_N) :-

(III.B) Nucleophilic addition reactions of $RMgX$ and RLi :-



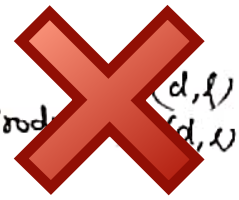
(i) Reactions with carbonyl compound :-

eg 4



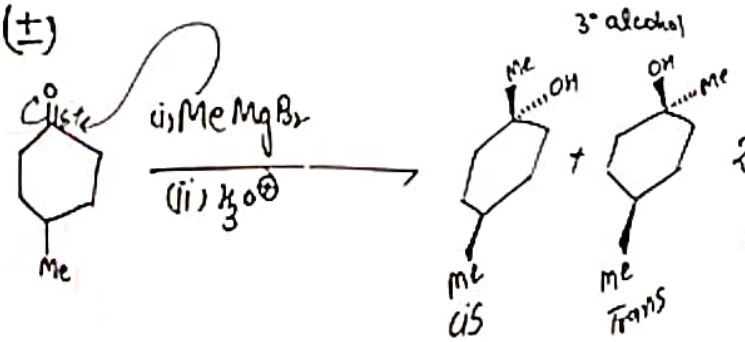
3° alcohol

(4 Prod (d,ℓ) (d,ℓ)



(±)

eg 5



+

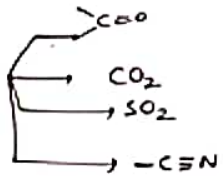
2. Geo. isomeric products

(optically inactive)

ORGANIC REACTION MECHANISM-I (ORM-I)

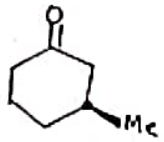
(III) Nucleophilic addition reactions (A_N) :-

(III.B) Nucleophilic addition reactions of $RMgX$ and RLi :-

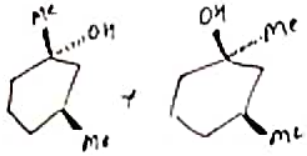


(i) Reactions with carbonyl compound :-

eg 6



(i) MeLi
(ii) $H_3O^{\oplus}$

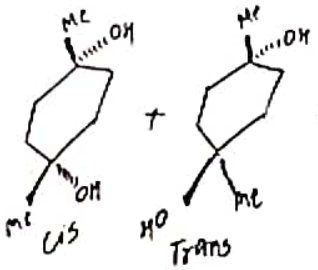


geo. isomers.
(diastereomers)
(optically active mix.)

eg 7



(i) MeMgBr (2 moles)
(ii) $H_3O^{\oplus}$

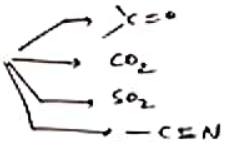


2. geo. isomers product
(Optically inactive)

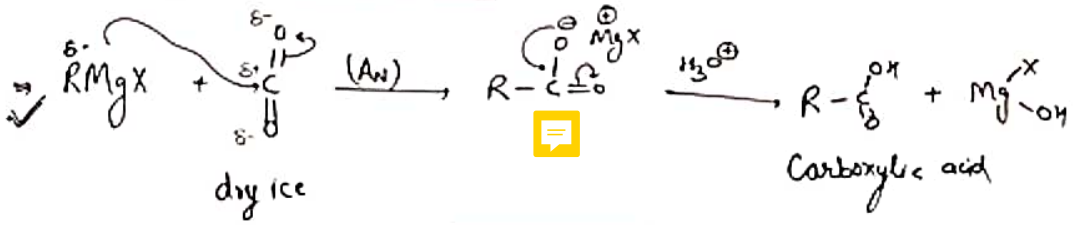
ORGANIC REACTION MECHANISM - I (ORM-I)

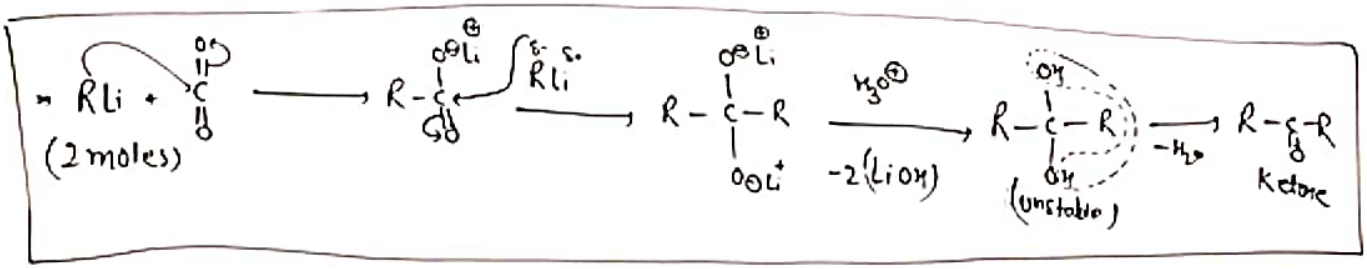
(III) Nucleophilic Addition Reactions (A_N) :-

(III.B) Nucleophilic addition reactions of RMgX & RLi



(B) Reaction with CO_2 / SO_2 / SO_3

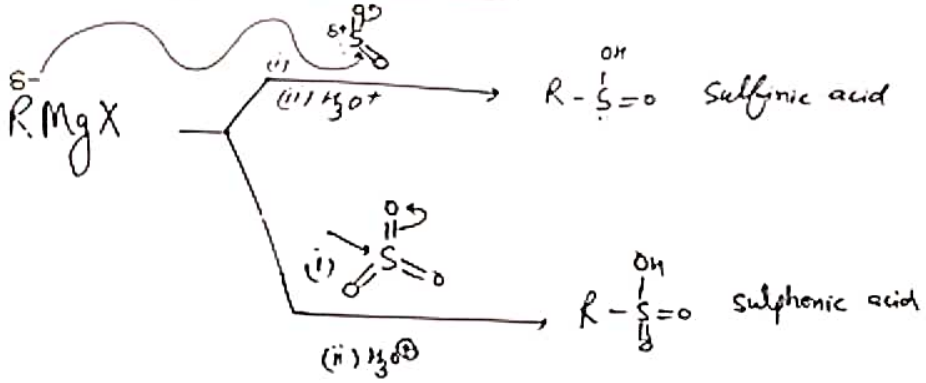




(III) Nucleophilic Addition Reactions (A_N) :-

(III.B) Nucleophilic addition reactions of RMgX & RLi → >C=O, CO_2, SO_2, $-C\equiv N$

(ii) Reaction with CO_2 / SO_2 / SO_3



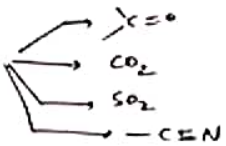
a (i) $PhMgX + CO_2 \xrightarrow[(ii)\ H_3O^{\oplus}]{} Ph\text{-}COOH$

(ii) $Ph\text{-}CH_2MgBr + CO_2 \xrightarrow[(ii)\ H_3O^{\oplus}]{} Ph\text{-}CH_2\text{-}COOH$

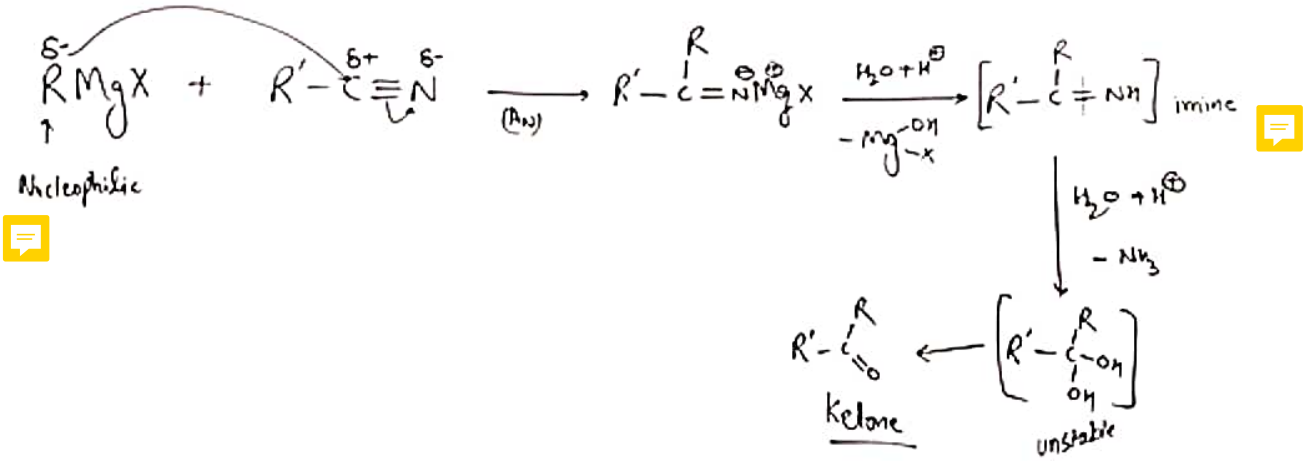
ORGANIC REACTION MECHANISM - I (ORM-I)

(III) Nucleophilic Addition Reactions (A_N) :-

(III. B) Nucleophilic addition reactions of RMgX & RLi



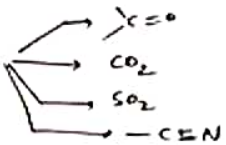
(iii) Reaction with R-C≡N :-



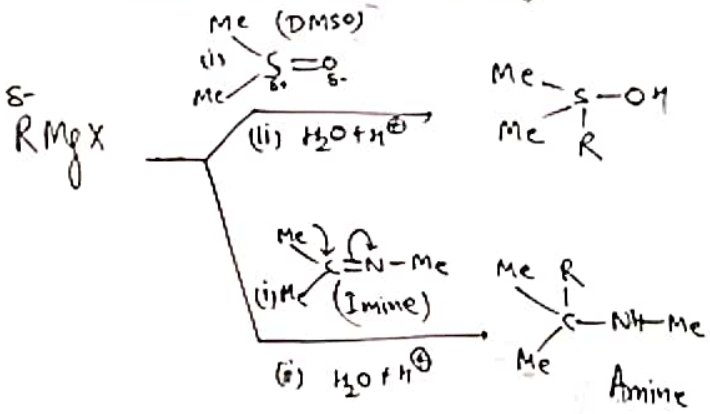
Nucleophilic

(III) Nuckophilic Addition Reactions (A_N) :-

(III B) Nucleophilic addition reactions of RMgX & RLi



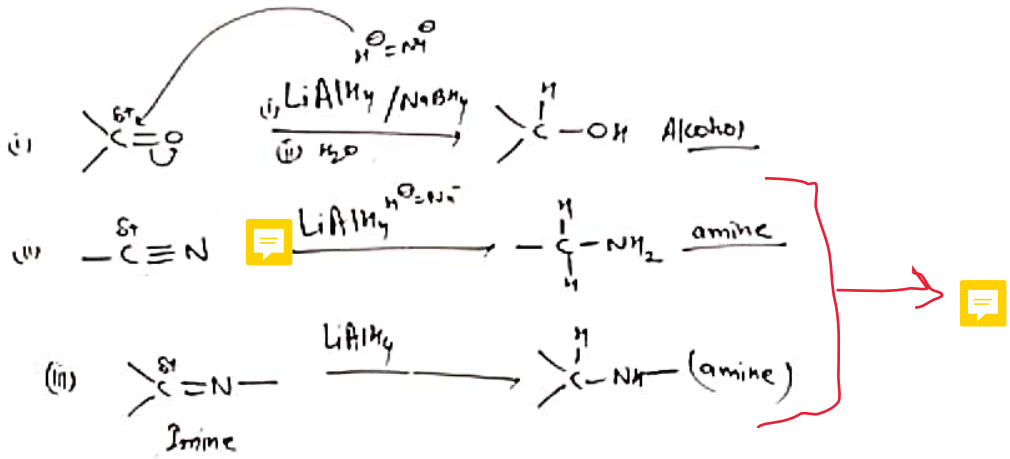
(IV) Other N_u Addition R_{x^n} of RMgX :-



ORGANIC REACTION MECHANISM - I (ORM-I)

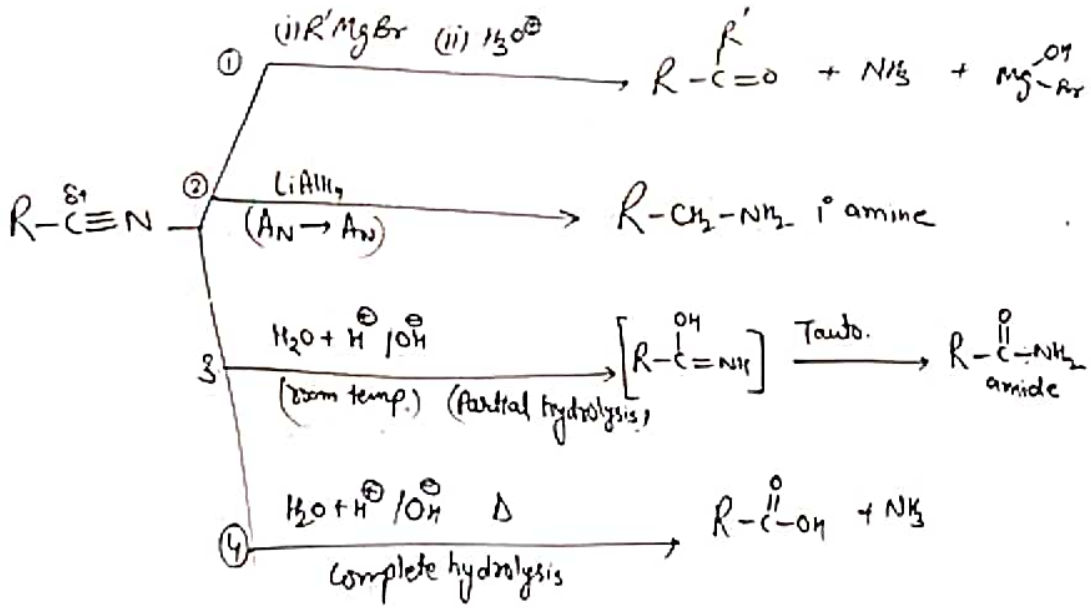
(III) Nuckophilic Addition Reactions (A_N) :-

(III C) Nucleophilic addition Rxn given by $LiAlH_4$ / $NaBH_4$



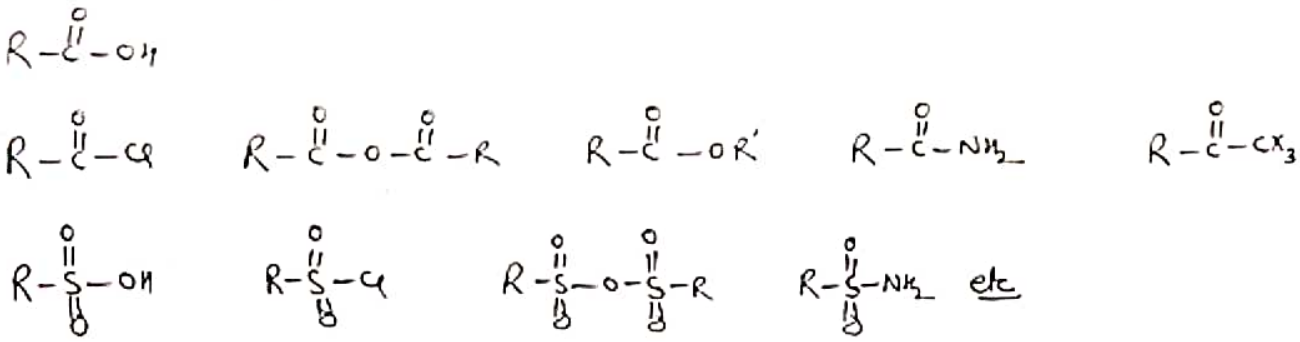
(III) Nucleophilic Addition Reactions (A_N) :-

(III.1) Nucleophilic addition Rxn of R–C≡N



(IV) S_N2Th = Acyl S_N2 Reaction

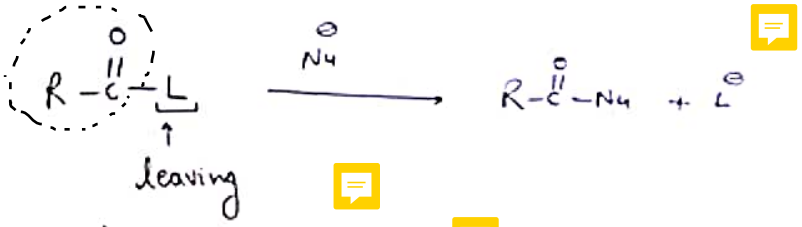
This reaction is given by acid and acid derivatives

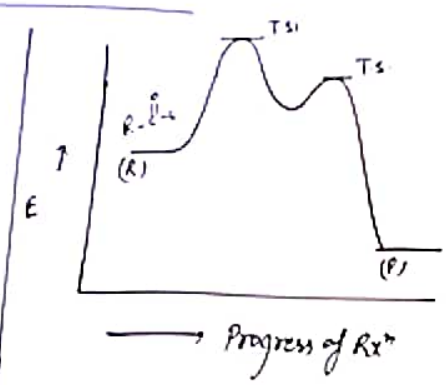


ORGANIC REACTION MECHANISM - I (ORM-I)

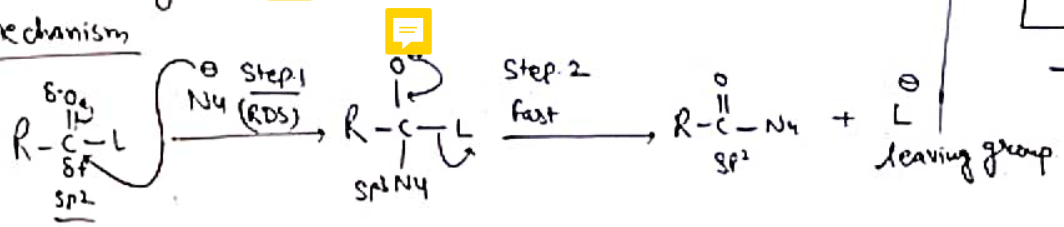
(IV) SN^2Th = Acyl SN^2 Reaction

- General mechanism for SN^2Th / Acyl SN^2 (Addition - elimination mechanism)





mechanism

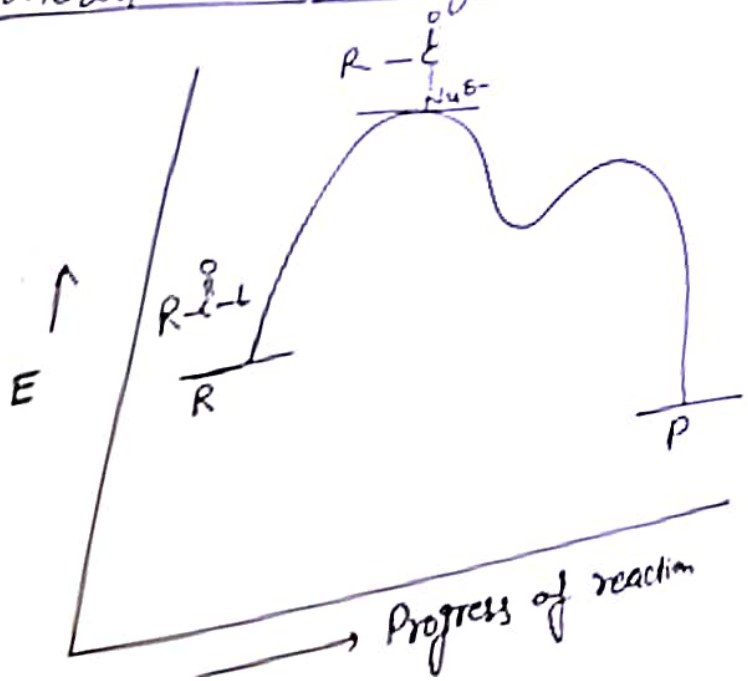


Tetrahedral intermediate

$\Rightarrow$ Rate of $SN^2Th \propto [R\text{-}C(=O)\text{-}L][N_4]$; 2nd order Rx^n

(IV) S_N^2Th = Acyl S_N^2 Reaction

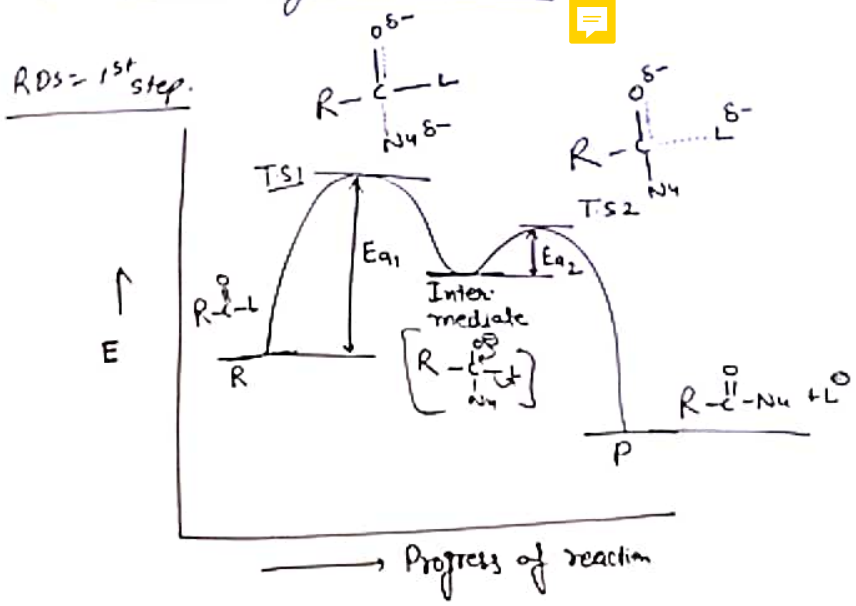
- General mechanism for $S_N^2\pi_h$/Acyl/S_N^2 (Addition-elimination mechanism)



ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) S_N^2Th = Acyl S_N^2 Reaction

RDS = 1st step.

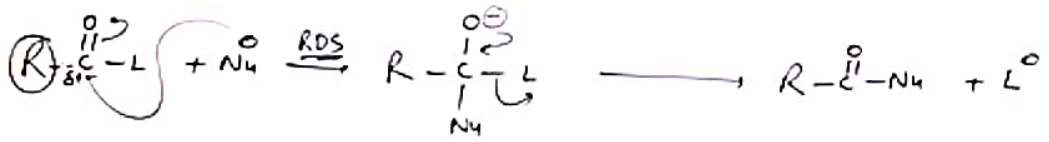


TS = Transition State

Ea = Activation energy

(IV) S_N^2Th = Acyl S_N^2 Reaction

Factors affecting reactivity of S_N^2Th reaction :-



(i) Reactivity of S_N^2Th $\propto$ Nucleophilicity of nucleophile.

(ii) " " " " $\propto$ magnitude of +ve charge of C of C=O group.

(iii) " " " " $\propto$ in presence of e^- withdrawing group of carbonyl group. $(-M, -I)$

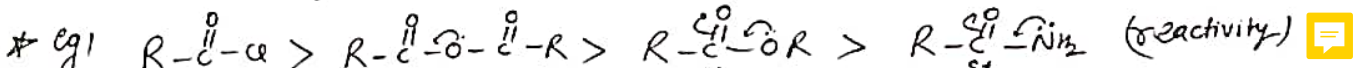
(iv) " " " $\propto$ " " " donating " " " " $(+M, +H, +I)$

(v) " " " $\propto \frac{1}{\text{Steric hindrance of C=O group}}$

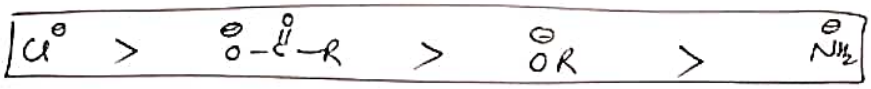
(vi) " " " $\propto$ Stability of leaving group

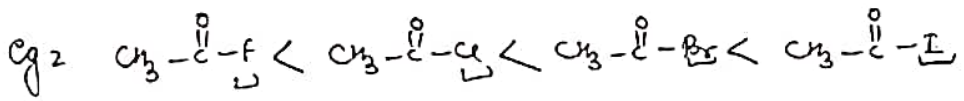
ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) S_N2Th = Acyl S_N2 Reaction

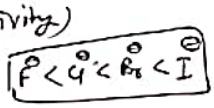
* eg 1 

(reactivity)

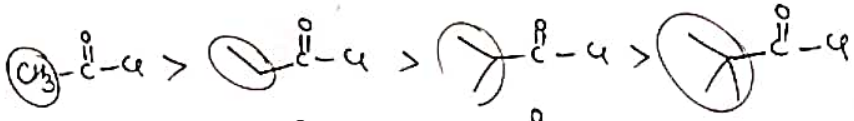
lg ability 

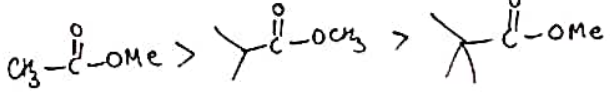
eg 2 

(reactivity)



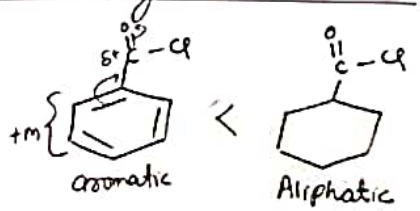
Stability

eg 3 

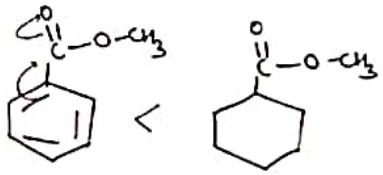
eg 4 

(IV) S_N^2Th = Acyl S_N^2 Reaction

eg 5

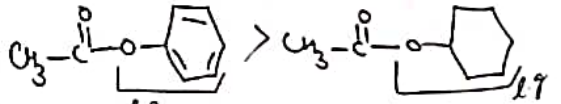


eg 6

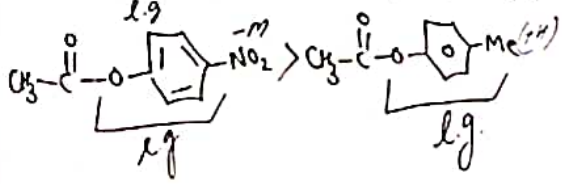


-O-R

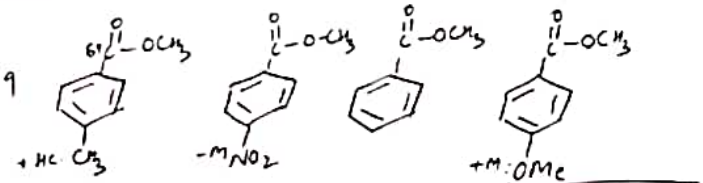
eg 7



eg 8

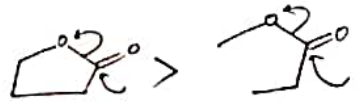


eg 9



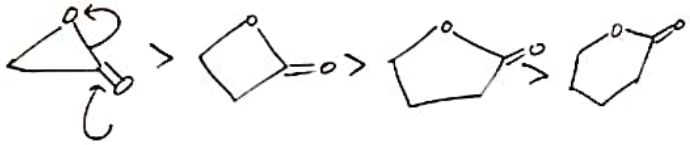
b > c > a > d

eg 10

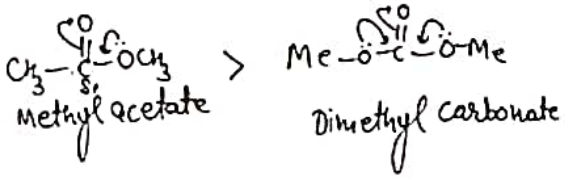


Cyclic Acyclic

eg 11



eg 12

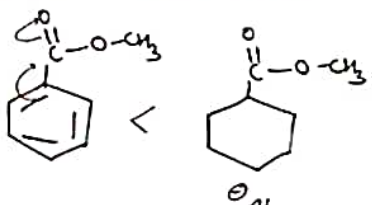


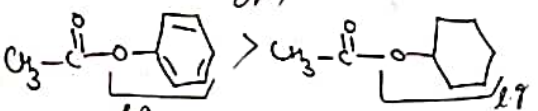
Methyl acetate Dimethyl carbonate

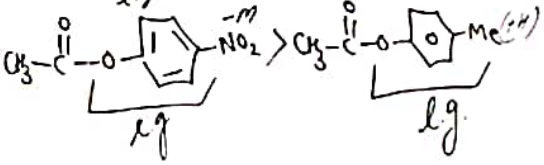
(IV) SN^2Th = Acyl SN^2 Reaction

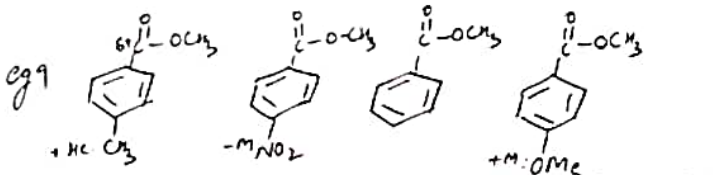
eg 13 $CH_3-\overset{O}{\overset{\|}{C}}-NH_2$ > $\ddot{N}H_2-\overset{O}{\overset{\|}{C}}-\ddot{N}H_2$

Acetamide

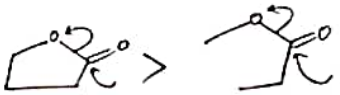
eg 6  $\ominus OPh$ $-O-R$

eg 7 

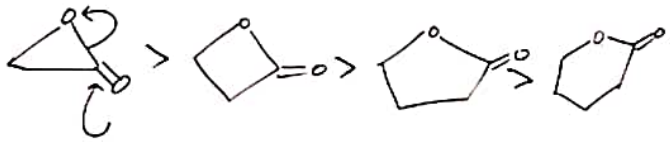
eg 8 

eg 9 

$b > c > a > d$

eg 10 

Cyclic Acyclic

eg 11 

eg 12 $CH_3-\overset{O}{\overset{\|}{C}}-OCH_3$ > $Me-O-\overset{O}{\overset{\|}{C}}-O-Me$

Methyl acetate Dimethyl carbonate

ORGANIC REACTION MECHANISM -I (ORM-I)

Reactions of SN^2Th mechanism:-

(I) General SN^2Th reactions of carboxylic acid derivative

(II) " " " " acid halide.

(III) Hydrolysis of acid derivatives

(IV) General SN^2Th reactions of carboxylic acid

(V) Esterification

(VI) SN^2Th reactive of PCl_5

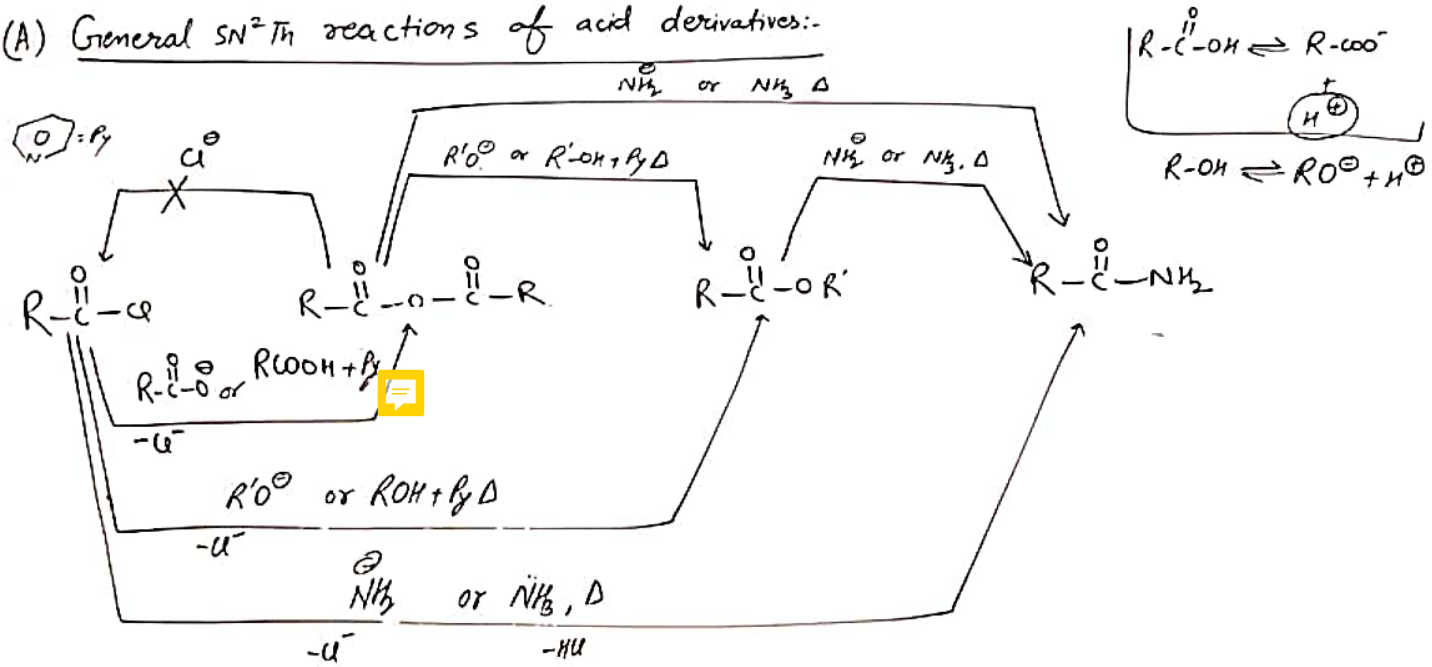
(VII) SN^2Th " " " Organometallic comp.

(VIII) " " " " LAH & $NaBH_4$

ORGANIC REACTION MECHANISM -I (ORM-I)

(IV) SN^2 Th Reactions:-

(A) General SN^2 Th reactions of acid derivatives:-

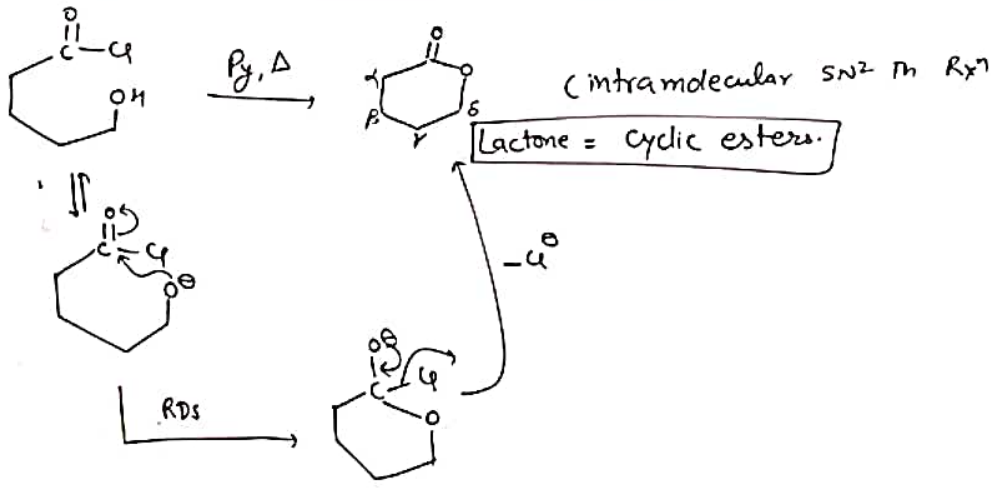


ORGANIC REACTION MECHANISM -I (ORM-I)

(IV) SN^2 Th Reactions:-

(A) General SN^2 Th reactions of acid derivatives:-

eg!



ORGANIC REACTION MECHANISM -I (ORM-I)

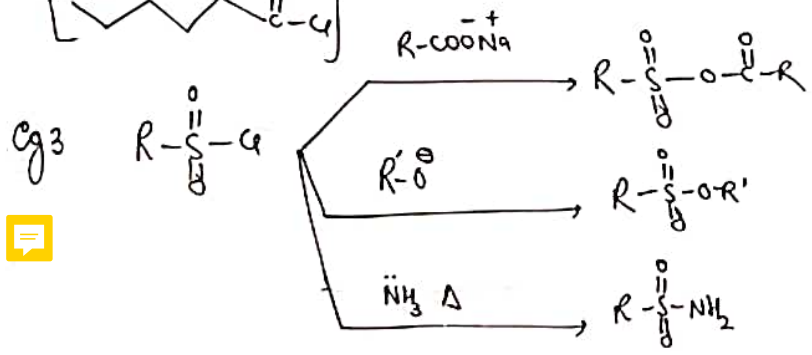
(IV) SN^2 Th Reactions:-

(A) General SN^2 Th reactions of acid derivatives:-

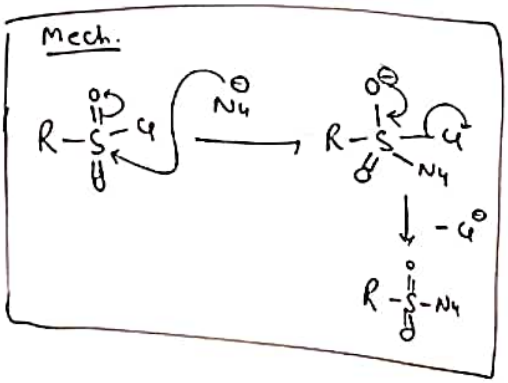
eg 2



eg 3



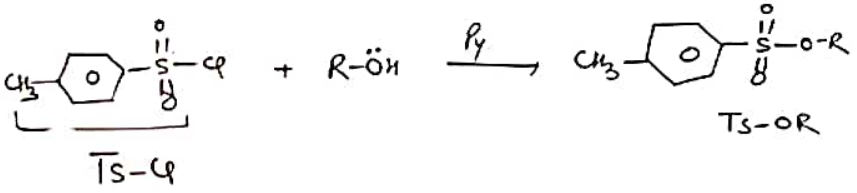
Mech.

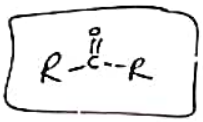


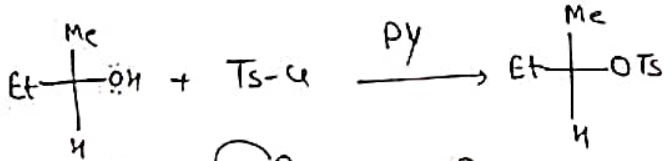
ORGANIC REACTION MECHANISM -I (ORM-I)

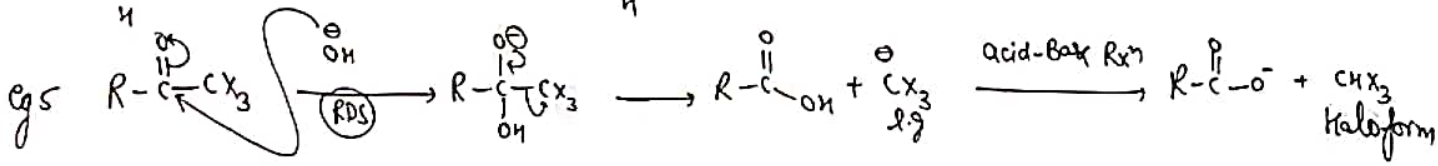
(IV) SN^2 Th Reactions:-

(A) General SN^2 Th reactions of acid derivatives:-

eg 4 



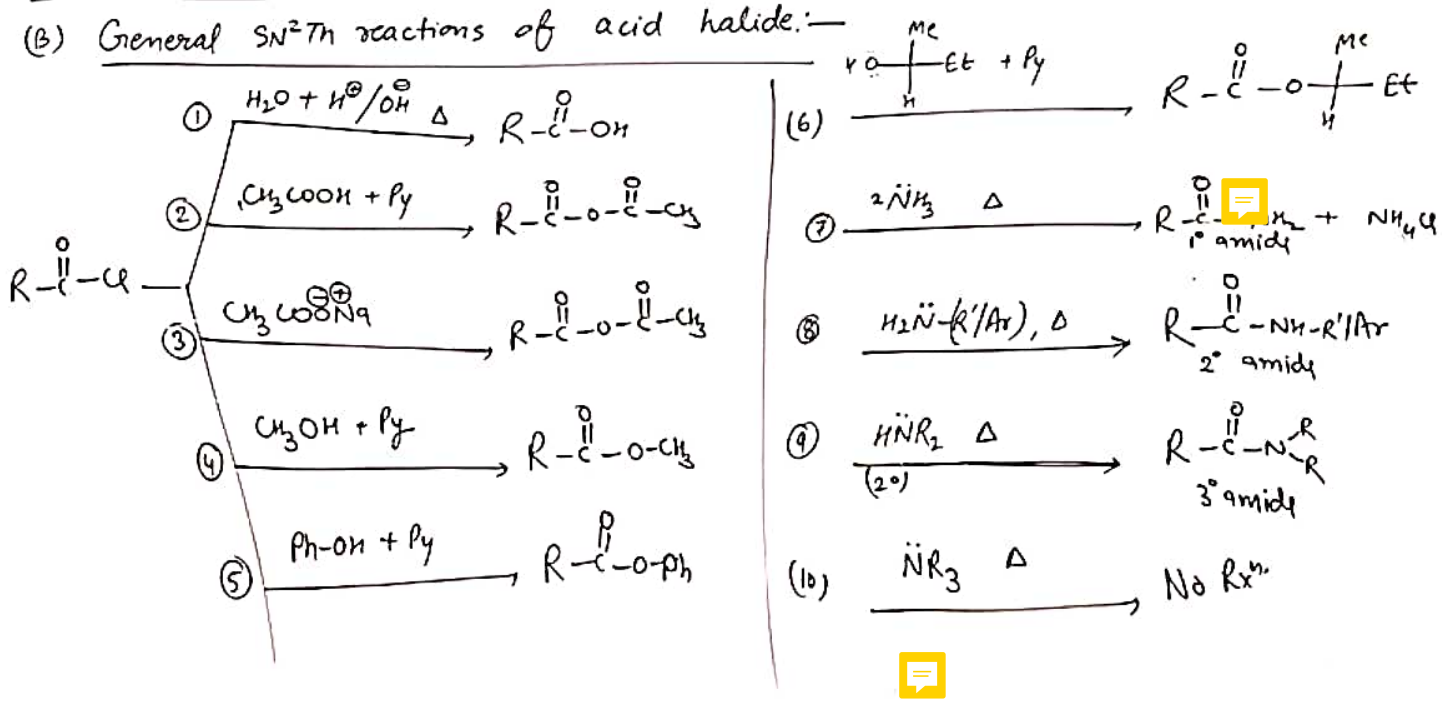
• 

eg 5 

ORGANIC REACTION MECHANISM -I (ORM-I)

(IV) SN^2 Th Reactions:-

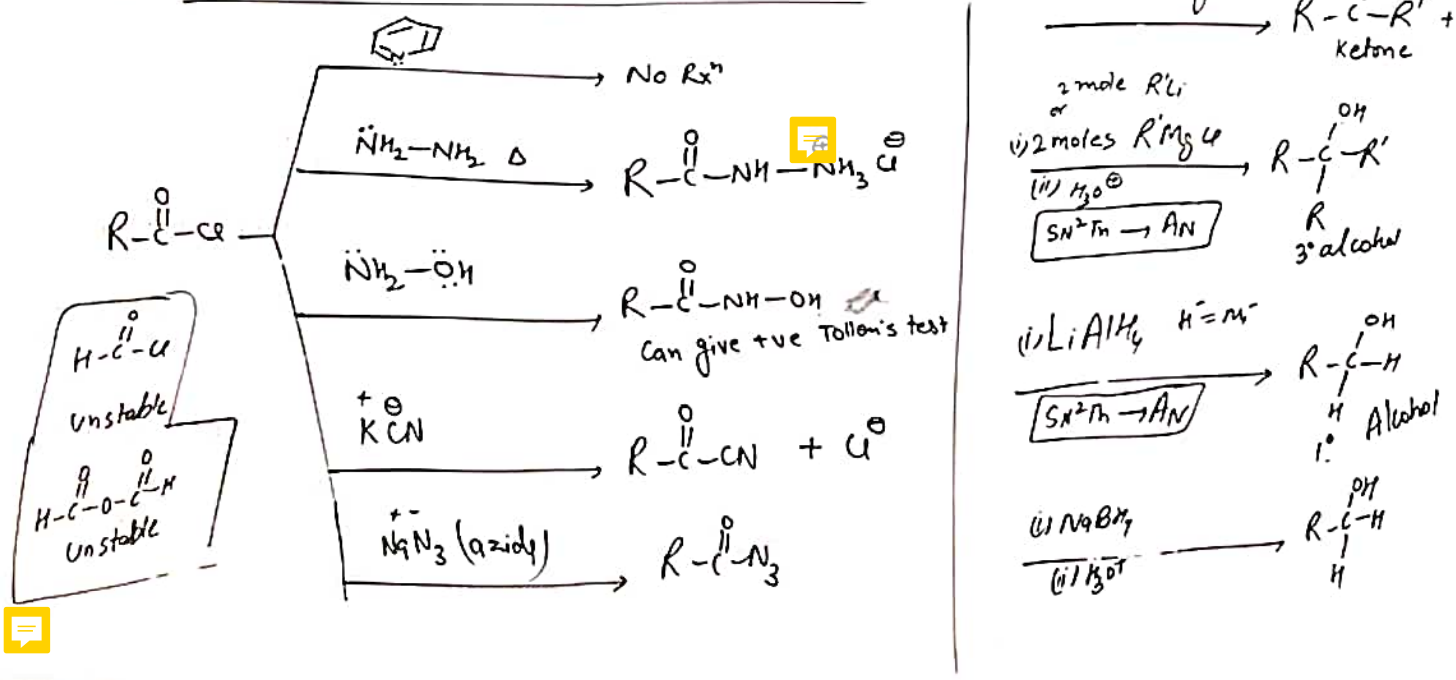
(B) General SN^2 Th reactions of acid halide:—

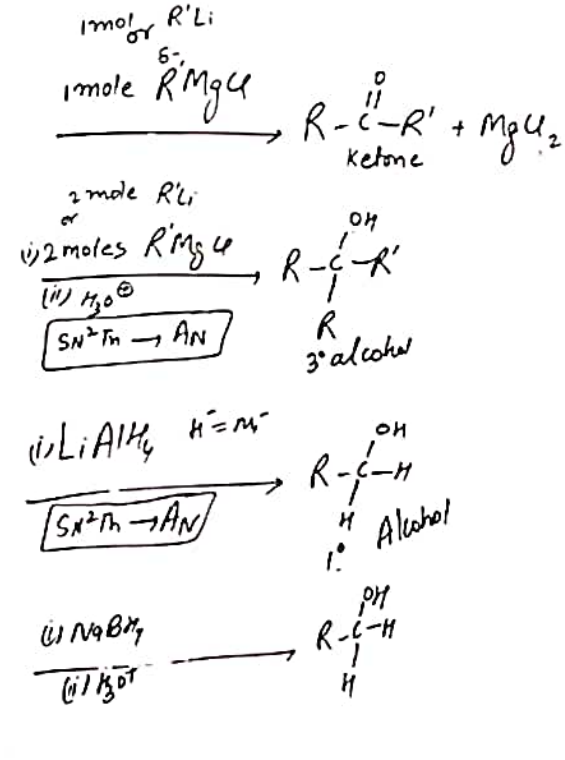


ORGANIC REACTION MECHANISM -I (ORM-I)

(IV) SN^2Th Reactions:-

(B) General SN^2Th reactions of acid halide:—

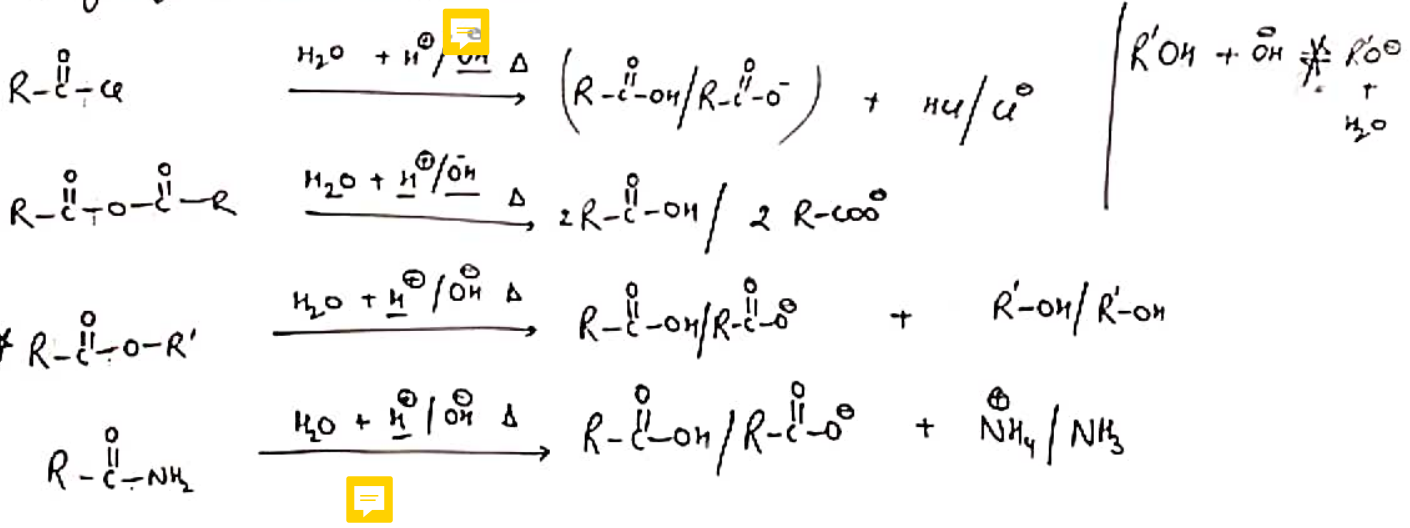




ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2 Th Reactions :-

(c) Hydrolysis of acid derivatives :-

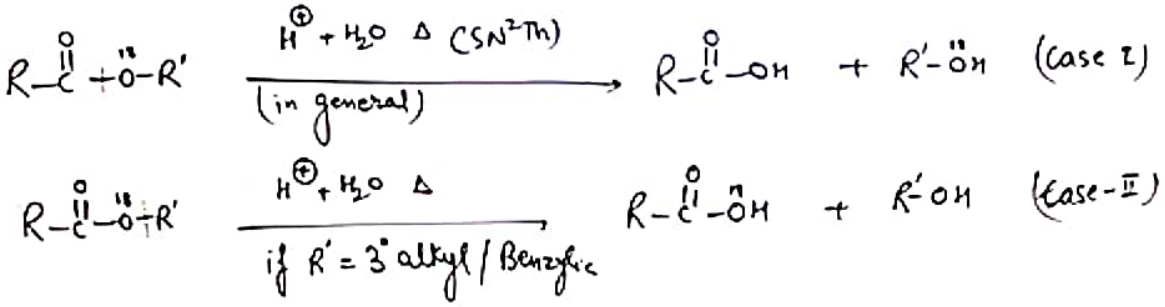


ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2Th Reactions :-

(c) Hydrolysis of acid derivatives :-

⇒ Hydrolysis of esters :-



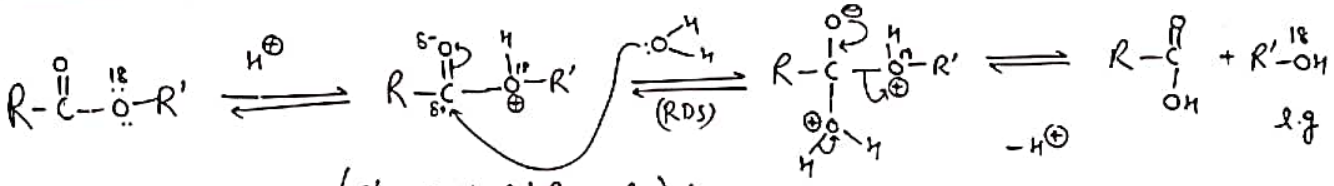
ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions:–

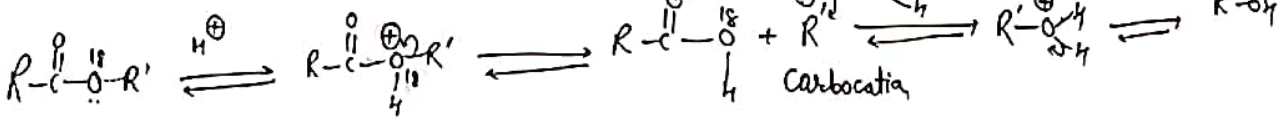
(c) Hydrolysis of acid derivatives :–

⇒ Hydrolysis of esters :–

mechanism for case I (SN^2 Th)



mechanism for case II (R' = 3° alkyl | Benzylic) (SN1)



ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions :-

(c) Hydrolysis of acid derivatives :-

⇒ Hydrolysis of esters :-

Note # Hydrolysis of esters in acidic medium ⇒ Reversible

" " " basic " " ⇒ Irreversible

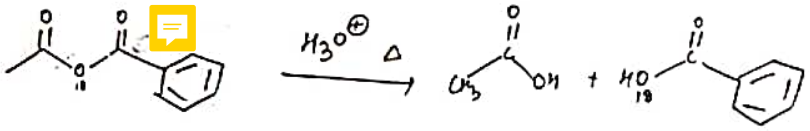
Hydrolysis of esters in basic medium = Saponification Rxn

ORGANIC REACTION MECHANISM – I (ORM-I)

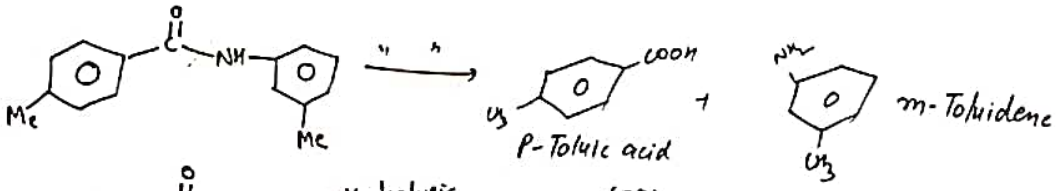
(IV) SN^2 Th Reactions :-

(c) Hydrolysis of acid derivatives :-

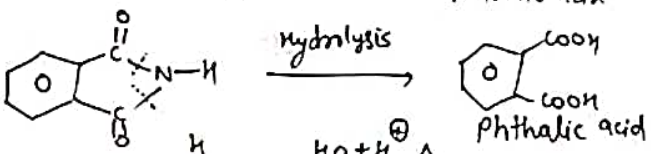
eg 1



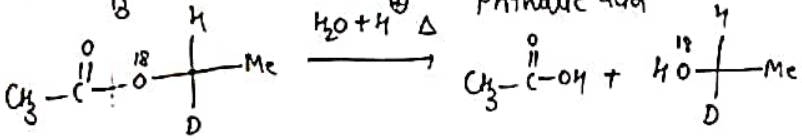
eg 2



eg 3



eg 4

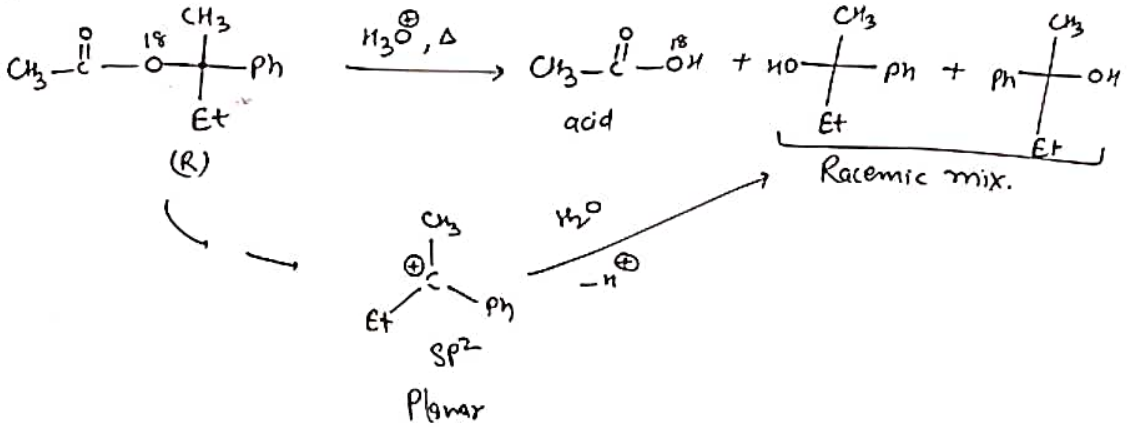


ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions:–

(c) Hydrolysis of acid derivatives :–

eg:

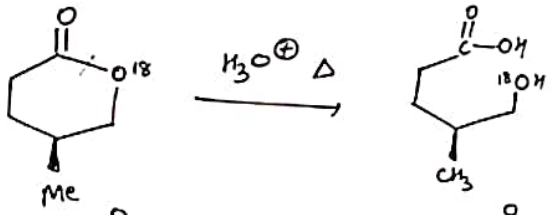


ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions :–

(c) Hydrolysis of acid derivatives :–

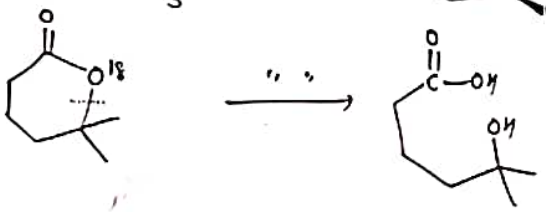
eg 6



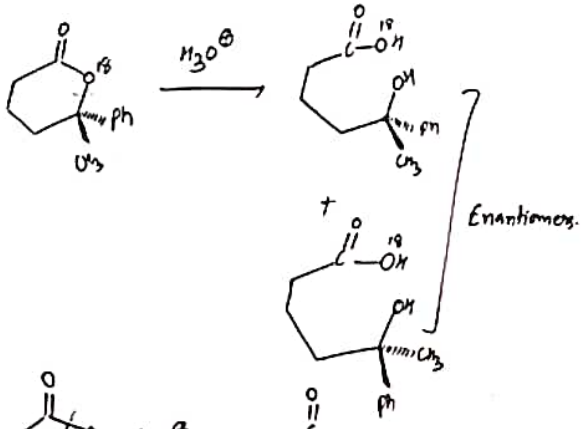
eg 7



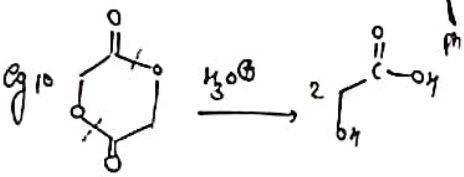
eg 8



eg 9



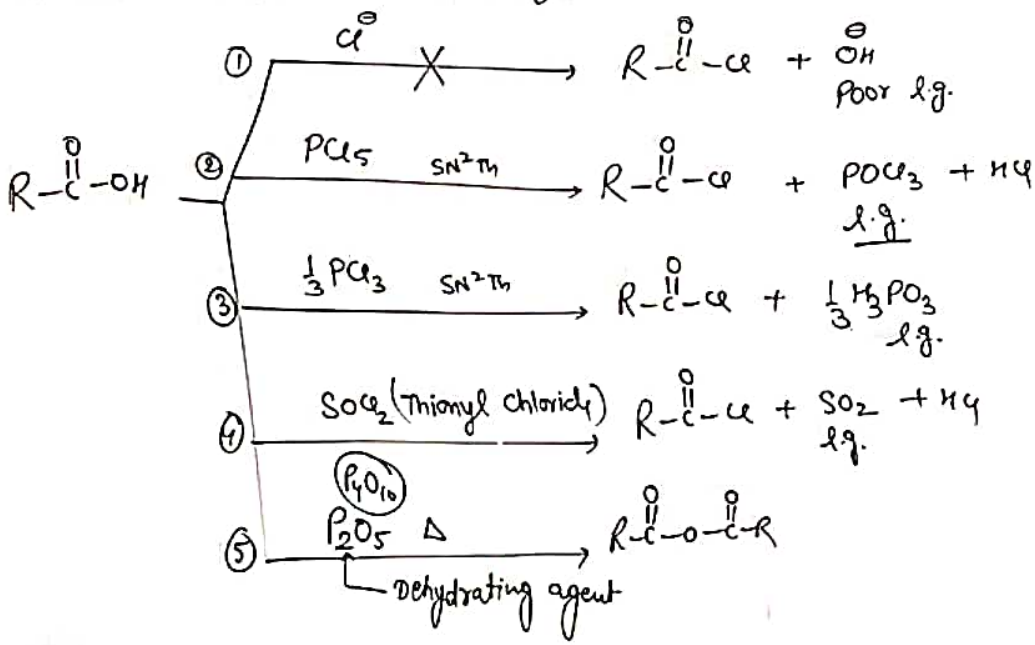
eg 10



ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions :-

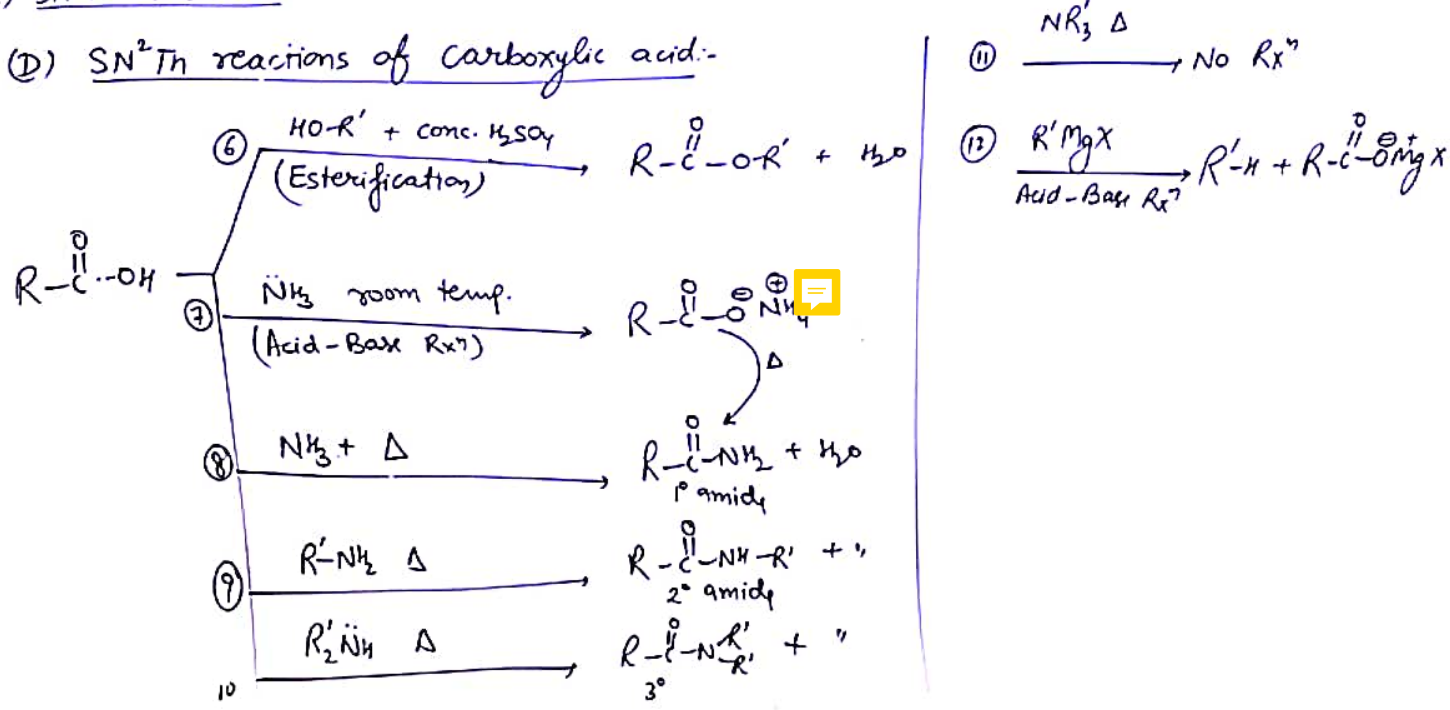
(D) SN^2 Th reactions of carboxylic acid :-



ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions:–

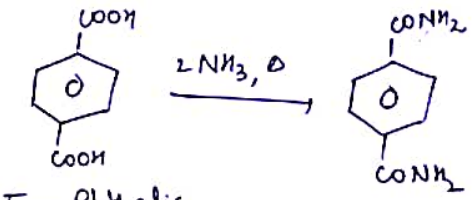
(D) SN^2 Th reactions of carboxylic acid:–



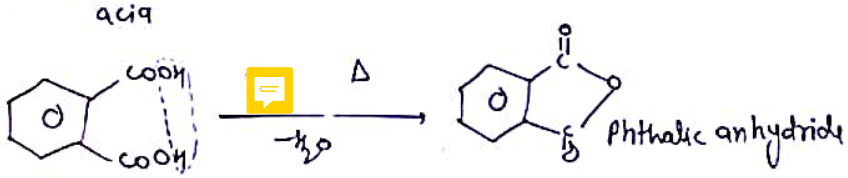
ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2Th Reactions:–

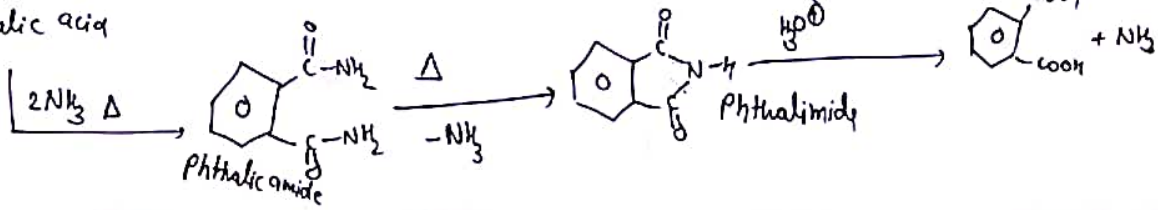
(D) SN^2Th reactions of carboxylic acid:-

eg 

Tere Phthalic acid

eg 

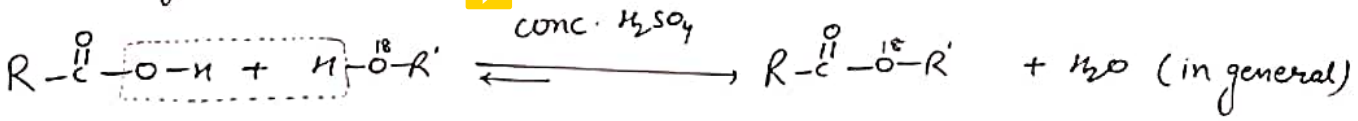
Phthalic acid

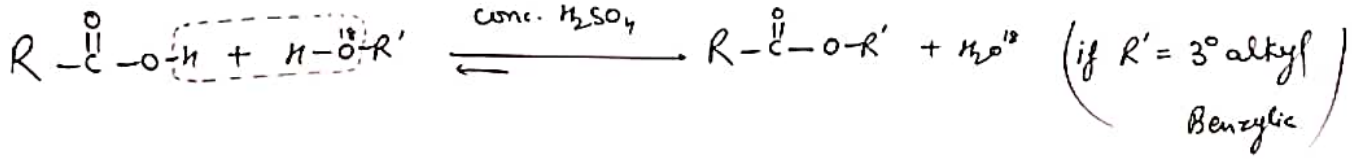


ORGANIC REACTION MECHANISM - I (ORM-I)

I) SN^2 Th Reactions:-

(E) Esterification:-

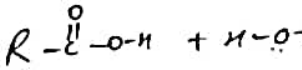


* 

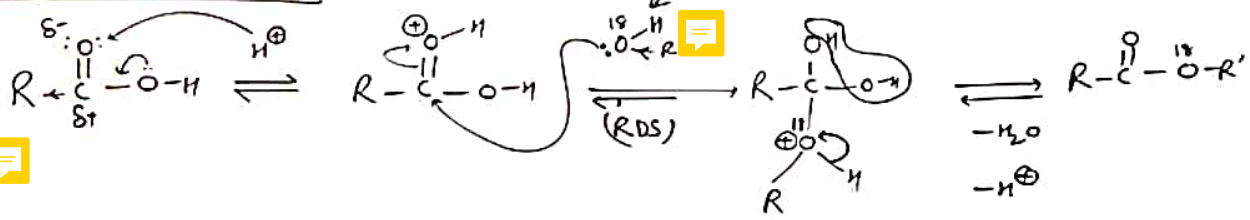
ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) S_N2 Th Reactions:–

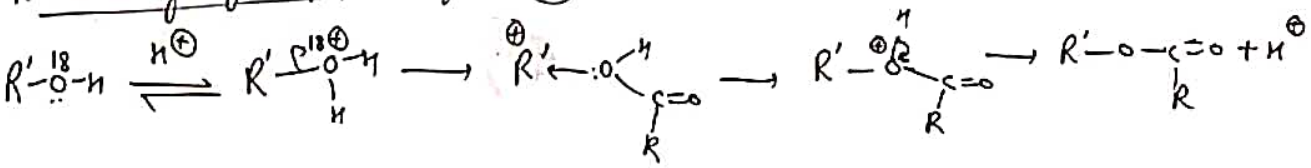
(E) Esterification:–



→ General mechanism



→ if R' = 3° alkyl group / Benzylic (S_N1) Rxn

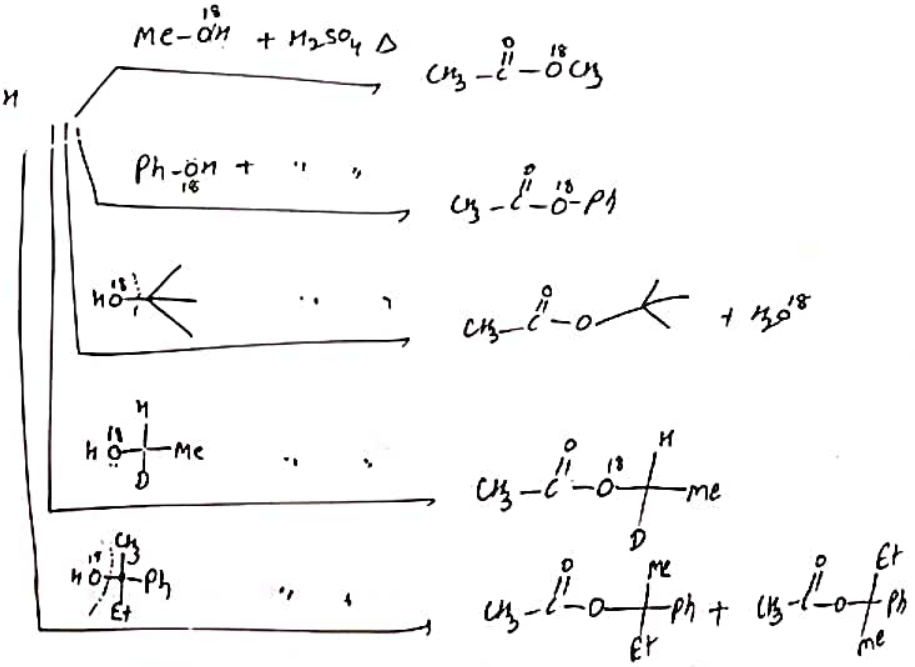


ORGANIC REACTION MECHANISM – I (ORM-I)

(IV) SN^2 Th Reactions:–

(E) Esterification:–

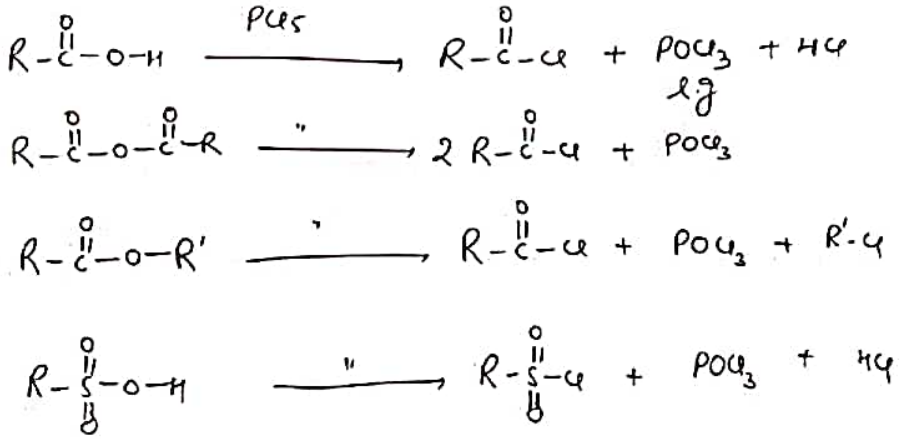
eg1 $CH_3-\overset{O}{\overset{\|}{C}}-OH$



ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2 Th REACTIONS :-

(F) SN^2 Th reaction, given by PCl_5

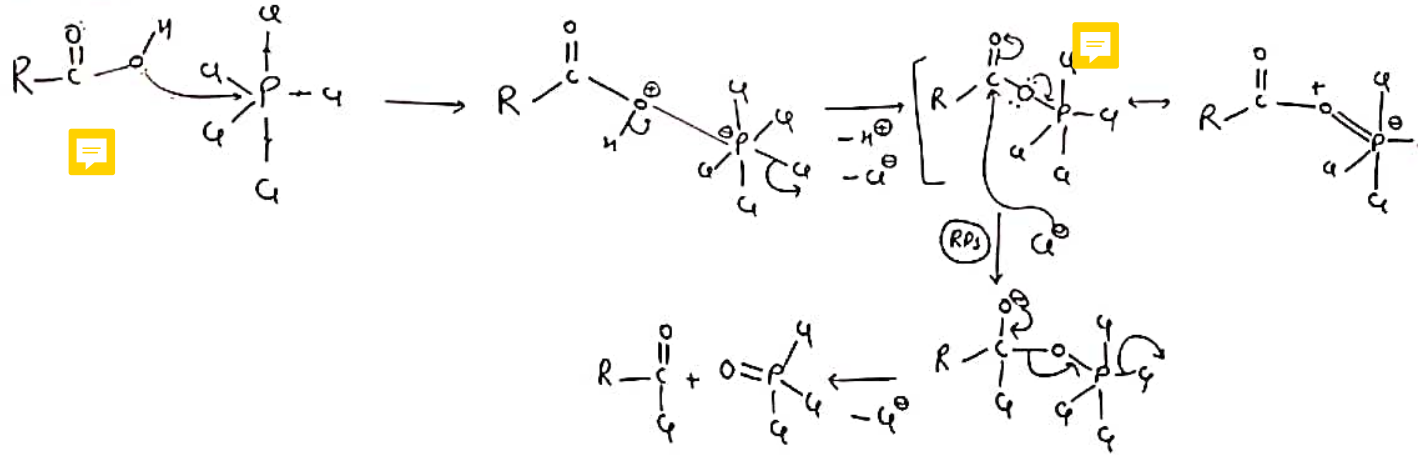


ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2Th REACTIONS :-

(F) SN^2Th reaction, given by PCl5

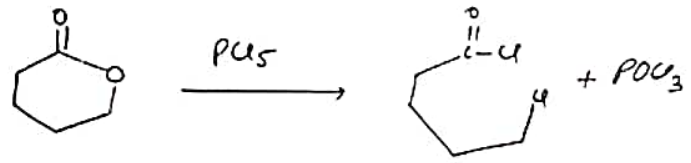
Mechanism

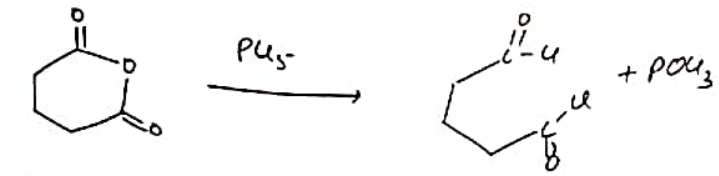


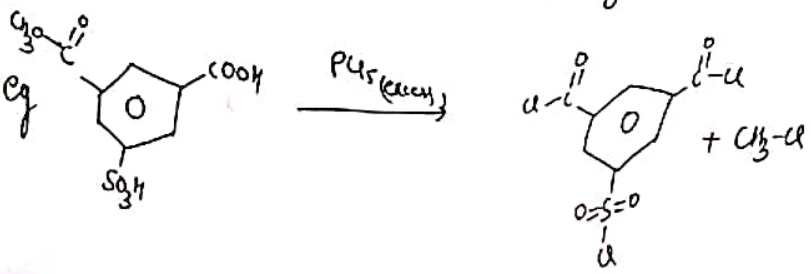
ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2Th REACTIONS :-

(F) SN^2Th reaction, given by PCl_5

eg 

eg 

eg 

ORGANIC REACTION MECHANISM - I (ORM - I)

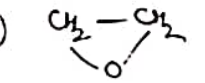
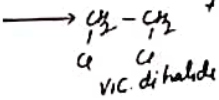
(IV) SN^2 Th REACTIONS :-

(F) SN^2 Th reaction, given by PCl_5

Other reactions of PCl_5 :-

① $R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl$

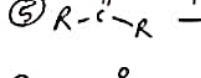
② $R-O-R + PCl_5 \longrightarrow 2R-Cl + POCl_3$

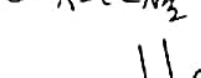
③  + ⋮ ⟶  + ⋮

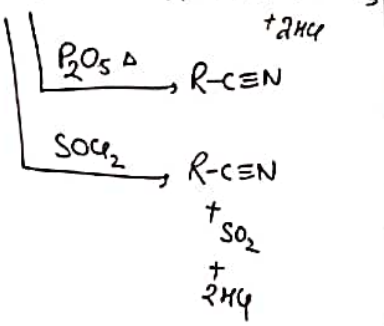
vic. dihalide

④ $R-CH=O + PCl_5 \longrightarrow R-CH(Cl)_2 + POCl_3$

Geminal dihalide

⑤ 

⑥ 



ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) S_N2 Th REACTIONS :-

(G) S_N2 Th reactions which are given by Organometallic compounds.
↳ Carbon and metal bond containing comp.

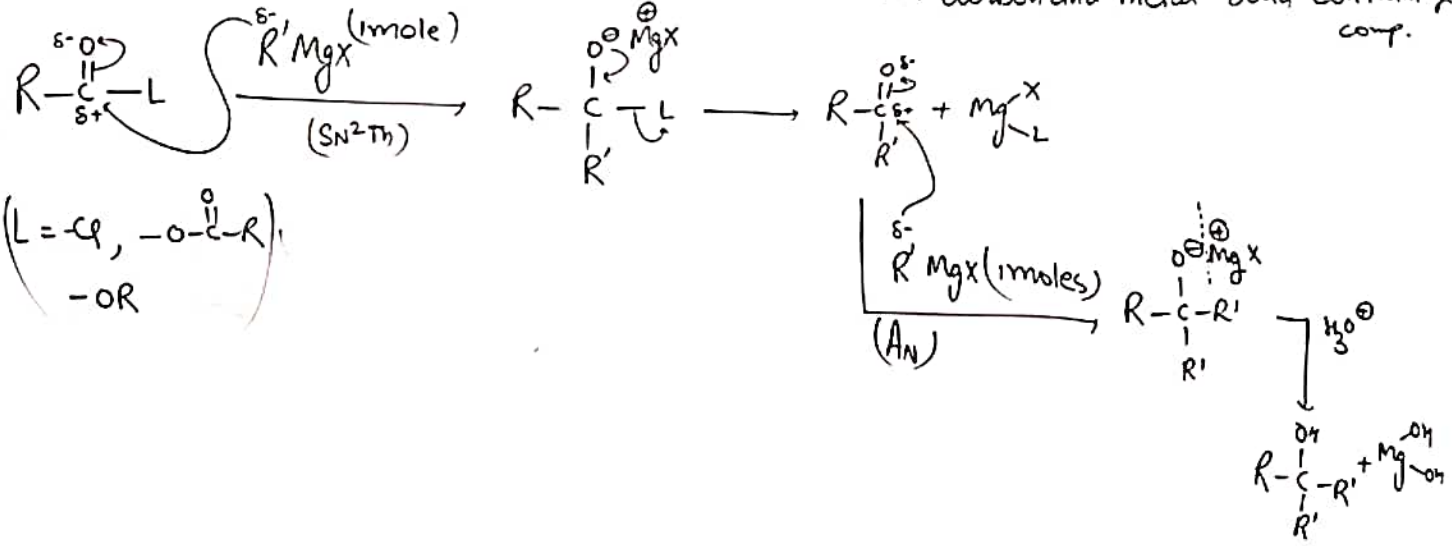
$-\overset{|}{\underset{|}{C}}^{\ominus}\ \overset{\oplus}{M}$	$-\overset{|}{\underset{|}{C}}^{\delta-}-M^{\delta+}$	$-\overset{|}{\underset{|}{C}}^{\delta\delta-}-M^{\delta\delta+}$
ionic		
M = Na, K	M = Mg, Li	M = Cd, Cd

ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.

↳ Carbon and metal bond containing comp.



(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.

Eg 1 $CH_3-\overset{O}{\overset{\|}{C}}-Cl$ + 1 mole $Et^{\delta-}MgBr$ → $CH_3-\overset{O}{\overset{\|}{C}}-Et$ (Ketone) + $Mg(Br)(Cl)$

(i) 2 moles $EtMgBr$
(ii) $H_2O + H^{\oplus}$ → $CH_3-\underset{Et}{\overset{OH}{\underset{|}{\overset{|}{C}}}}-Et$ (3° alcohol)

Eg 2 $H-\overset{O}{\overset{\|}{C}}-O-Ph$ (Phenyl formate) + 1 mole $MeMgBr$ → $H-\underset{Me}{\overset{O}{\underset{|}{\overset{\|}{C}}}}$ (Ald.) + $Ph-O^{-}Mg^{+}Br$

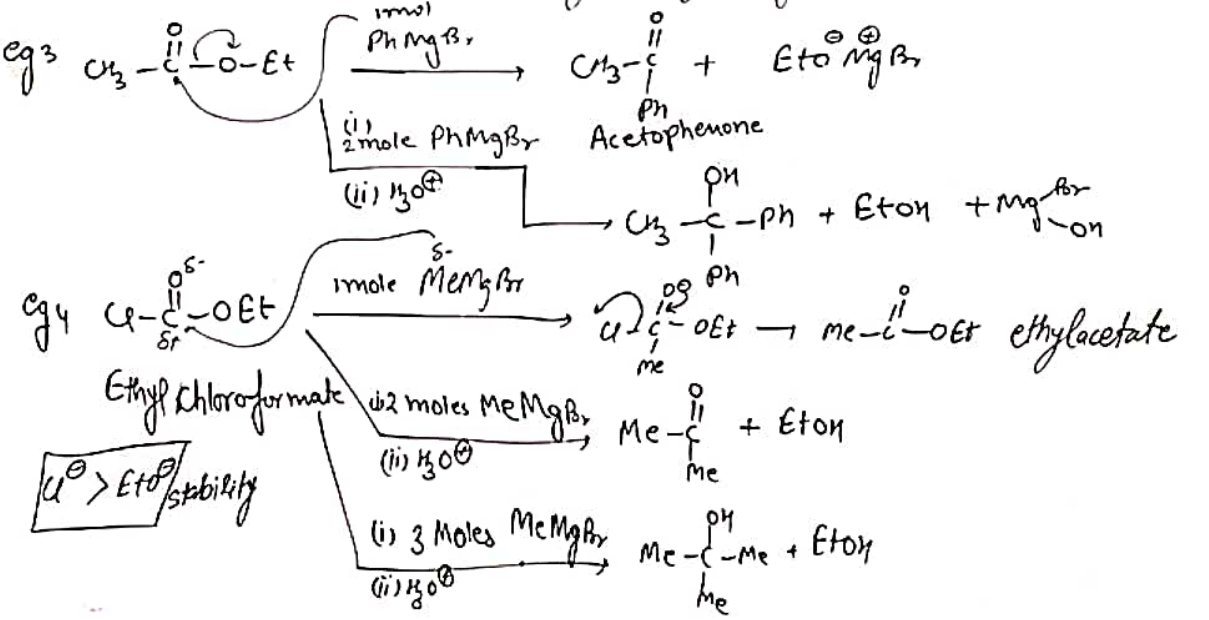
2 moles $MeMgBr$ ($SN^2Th \rightarrow AN$) → $H-\underset{Me}{\overset{O^{\ominus}MgBr}{\underset{|}{\overset{|}{C}}}}-Me$ + $Ph-O^{-}MgBr$

$H_3O^{\oplus}$ → $H-\underset{Me}{\overset{OH}{\underset{|}{\overset{|}{C}}}}-Me$ (2° alcoh.) + $Ph-OH$ (Phenol) + $2\,Mg(Br)(OH)$

ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2Th REACTIONS :-

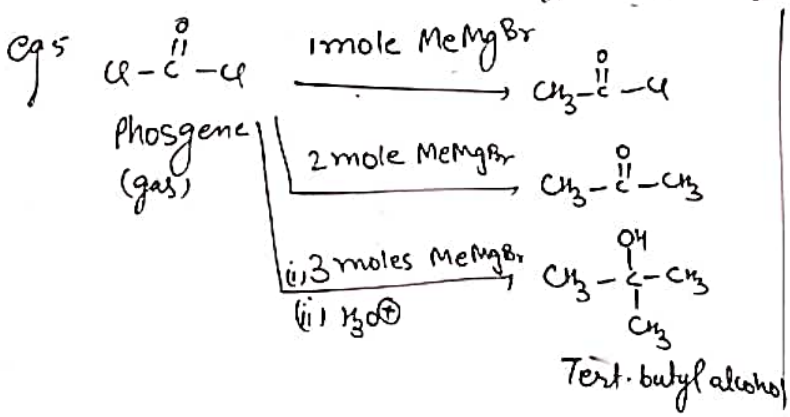
(G) SN^2Th reactions which are given by Organometallic compounds.

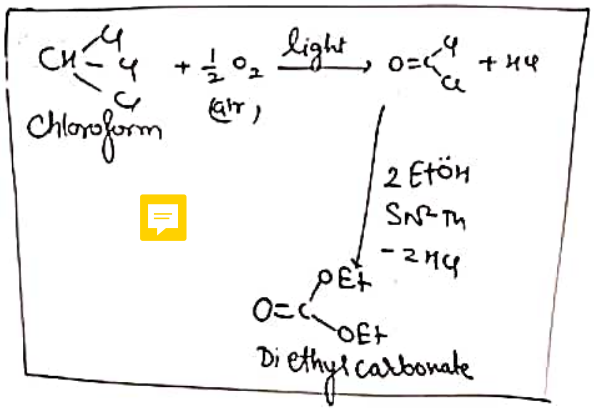


ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.



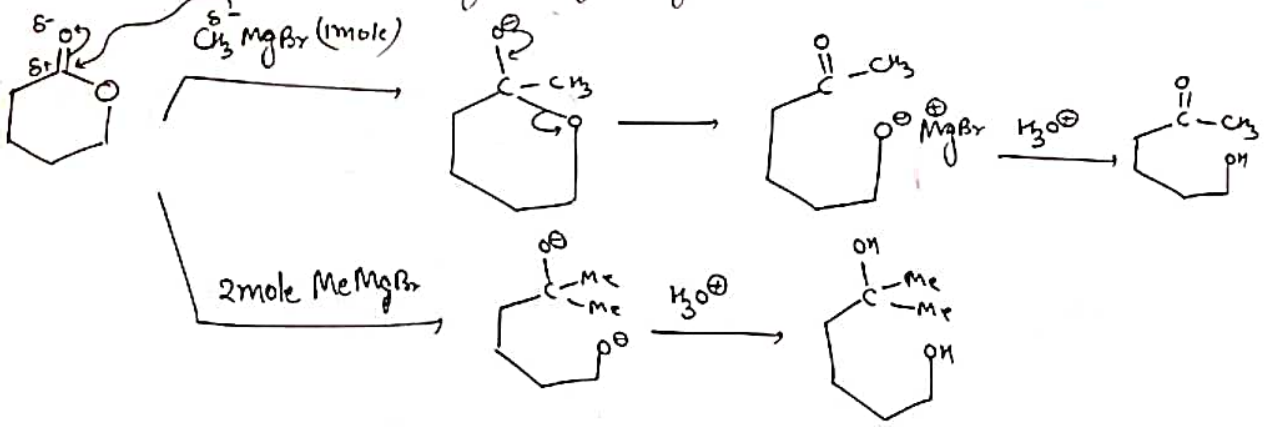


ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.

eg 6

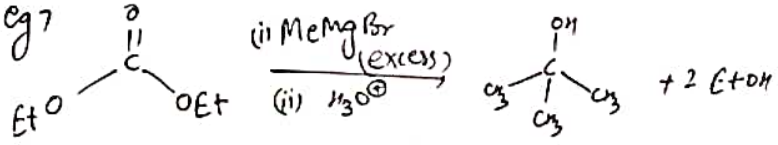


ORGANIC REACTION MECHANISM - I (ORM - I)

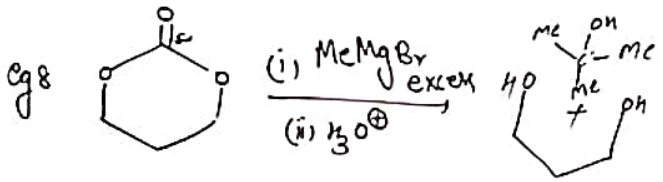
(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.

Eg 7



Eg 8



ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2 Th REACTIONS :-

(G) SN^2 Th reactions which are given by Organometallic compounds.

Q ester $\xrightarrow[\text{excess}]{\text{(i) MeMgBr}}$ $\xrightarrow[\text{(ii) } H_3O^{\oplus}]{}$ isopropyl alcohol

?

(Only single org. comp.)

Sol. $R-C(=O)-O-R' \xrightarrow{MeMgBr} R-C(OH)(Me)-Me + R'OH$

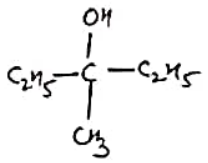
$(R = H)$ $R' = CH_3-CH(CH_3)-$

Ans $H-C(=O)-O-CH(CH_3)_2$

ORGANIC REACTION MECHANISM - I (ORM-I)

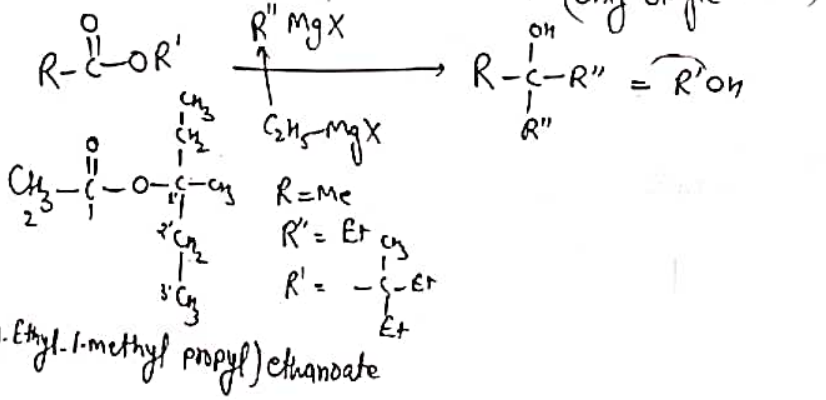
(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.

Q ester $\xrightarrow[\text{(ii) } H_3O^{\oplus}]{RMgX}$ 

? = IUPAC Name

(only single alcohol)

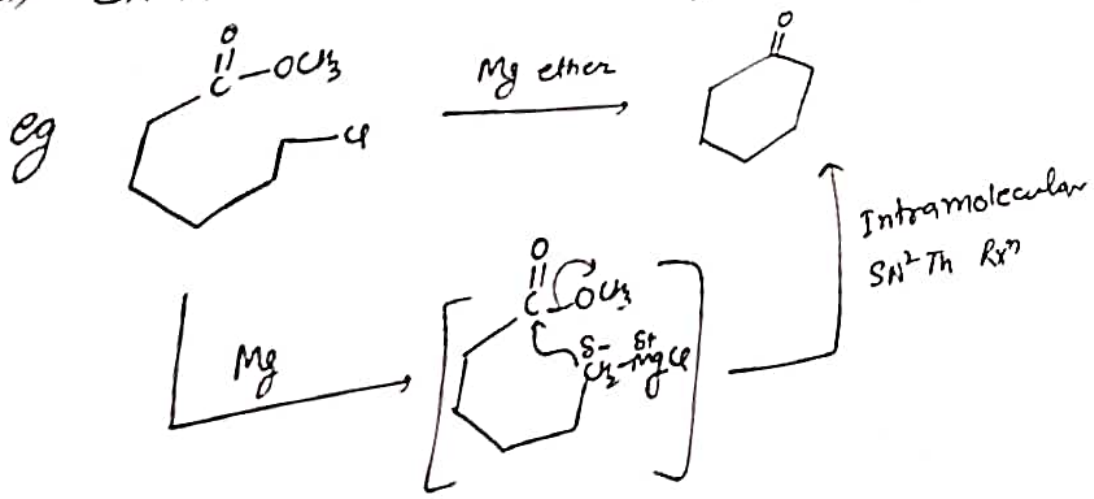
Sol. 

(1-Ethyl-1-methyl propyl) ethanoate

ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2Th REACTIONS :-

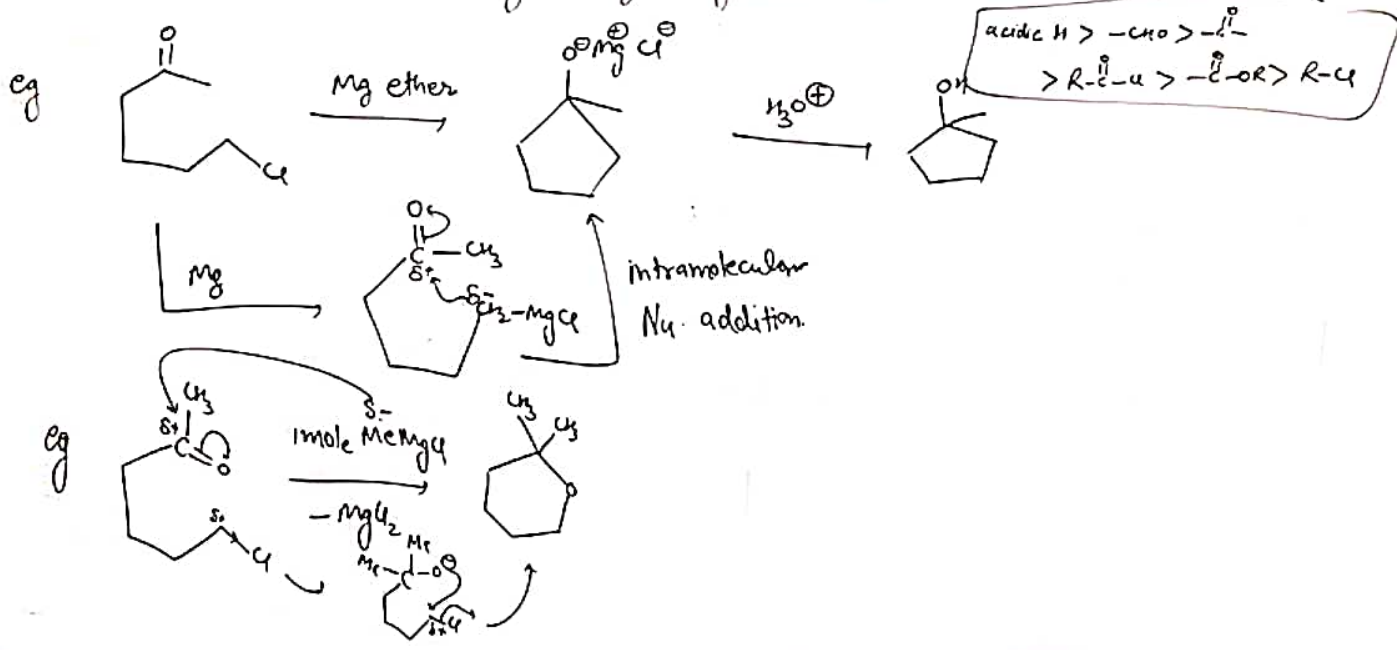
(G) SN^2Th reactions which are given by Organometallic compounds



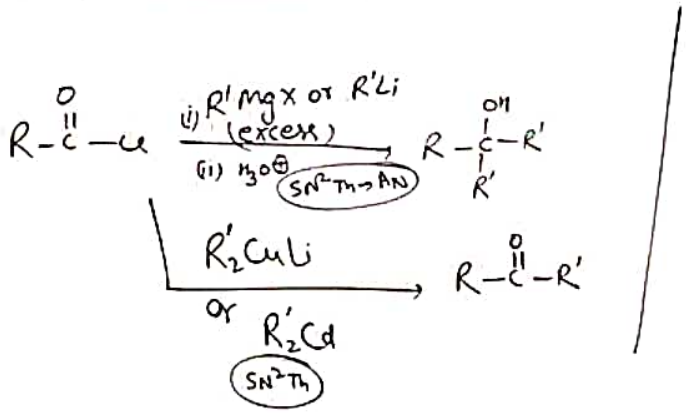
ORGANIC REACTION MECHANISM - I (ORM-I)

(IV) SN^2Th REACTIONS :-

(G) SN^2Th reactions which are given by Organometallic compounds.



(IV) SN^2Th Reactions :-



R_2CuLi and R_2Cd ⇒ No Reaction with $-\overset{O}{\overset{\|}{C}}-H$, $-\overset{O}{\overset{\|}{C}}-R$, $-C\equiv N$ etc.

↑ weaker than RMgX

(IV) SN^2 Th REACTIONS :-

Formation of R_2CuLi

$$R-X + 2Li \xrightarrow{\text{ether}} RLi + LiX$$

$$2RLi + CuI \longrightarrow R_2CuLi + LiI$$

⇒ Formation of R_2Cd

$$2RMgCl + CdCl_2 \longrightarrow R_2Cd + 2MgCl_2$$

ORGANIC REACTION MECHANISM - I (ORM - I)

(IV) SN^2Th REACTIONS :-

(H) SN^2Th reactions which are given by $LiAlH_4$ $NaBH_4$ $H^{\ominus}$

